

Gradual weakening across morphological layers:

Solving stratal paradoxes in Kashaya

Eva Zimmermann, Leipzig University, Eva.Zimmermann@uni-leipzig.de

August 2025, submitted manuscript

Abstract: Harmonic Layer Theory can correctly predict the interaction of metrical processes in Kashaya which results in an ordering paradox in a classical stratal system. This novel cyclic formalization of the morpho-phonology interface predicts that processes can be restricted to elements that already underwent a previous optimization cycle – while novel elements added in a later cycle are simultaneously present.

Keywords: Kashaya, Harmonic Layer Theory, Strata, Opacity, Iambic Lengthening

1 Introduction

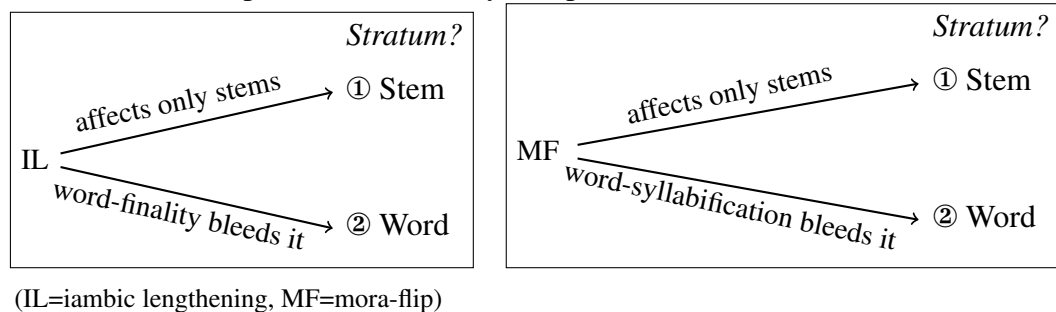
The metrical system of Kashaya is a challenge for any restrictive model of the morpho-phonology interface. On the one hand, there is converging evidence for a classical stratal distinction. Certain processes like iambic lengthening only affect roots and stem-level affixes but never more peripheral word-level affixes. At the same time, such processes which should seemingly be part of an early stem-stratum phonology can be fed or bled by processes which clearly apply at the word-level. Assigning operations to either a stem- or word-stratum hence fails to explain the interaction of metrical processes in Kashaya. In Buckley (2017), this ordering paradox of a global effect across lexical strata is illustrated with a detailed discussion of the additional assumptions necessary to predict the Kashaya pattern in various cyclic and non-cyclic theories.

This paper has three goals: Firstly, it argues for a novel theoretical account of the metrical system of Kashaya within Harmonic Layer Theory. It can hence be understood

as a reply to the statement that ‘[d]epending on what compromises one is willing to make, the challenge might not seem especially grave’ made in Buckley (2017) and claims that the price one needs to pay in Harmonic Layer Theory is smaller and has empirically more adequate consequences than the price one needs to pay in alternative accounts. This novel account aims to incorporate the best of both a cyclic account and of a parallel OT-account: It is a cyclic model that explains opacity via reference to intermediate forms but is still able to predict conspiracies across strata from a single constraint grammar. It hence dispenses with foot extrametricality or co-grammars, which are necessary in existing alternative accounts of Kashaya. Secondly, this paper presents a general argument for Harmonic Layer Theory, a novel architecture of the morpho-phonology interface that relies on cyclic optimizations with a single grammar. Whereas previous arguments for Harmonic Layer Theory emphasize that this model is more restrictive than a stratal model based on different co-grammars for each stratum, the Kashaya case study emphasizes a qualitative difference between both models. Harmonic Layer Theory does not assign processes to specific strata, which makes substantially different predictions about the possible interactions of processes than a classical stratal system. As this paper argues, this difference between the systems makes the Kashaya patterns easily predictable under Harmonic Layer Theory but runs into complications in a stratal system. Thirdly, this paper wants to re-emphasize the relevance of detailed case studies like Kashaya for any discussion of the morpho-phonology interface. That the complex interaction of processes is challenging under various theoretical assumptions has been discussed extensively over the years, in particular in the work of Eugene Buckley. These detailed empirical and theoretical investigations are highly relevant beyond the specific Kashaya pattern: They show us that theoretical models need not only to strive to predict certain typological tendencies or challenging isolated phenomena, they also need to be empirically adequate to handle complex

case studies of a single language that take into account multiple interacting aspects of the morpho-phonological system. The arguments in this paper are based exclusively on existing empirical descriptions and the theoretical generalizations established in existing work but differ nevertheless with respect to the formal explanations offered for a specific empirical pattern. The paper hence also emphasizes the importance of distinguishing between empirical surface facts and the generalizations predicted by a theory – the former can be identical but the latter can differ between analyses. The depiction (1) gives an outlook of the two most crucial ordering paradoxes in Kashaya. Both iambic lengthening and a process called mora-flip affect only stem material and never word-level affixes. At the same time, the correct context of application for both processes can only be determined at the word level: Iambic lengthening never applies to word-final vowels and word-level syllabification can bleed mora-flip.

(1) Order of metrical processes in Kashaya: A paradox



I argue that these ordering paradoxes fall out in an account within Harmonic Layer Theory which is based on a simple change of perspective: Processes do not apply at certain strata or optimization cycles, they apply to material that was optimized a certain number of times. Since morphemes enter the derivation at different points in a cyclic model, there might hence be a heterogeneous structure where only a subset of material is prone to a certain process. More concretely, iambic lengthening and mora-flip in Kashaya only apply to vowels that were already optimized once and thus at the second ($\sim$ word level) optimization. Crucially, iambic lengthening and mora-flip thus apply to stem material while

word-level material is already present. Word-level material remains unaffected since it is only in its first optimization cycle but its presence might nevertheless feed or bleed certain processes. This is schematized in (2). Such an account where mora-flip is a late process easily predicts a third peculiar property of the Kashaya metrical system, namely, opaque stress shift whose context is destroyed by mora-flip. In a cyclic model, this becomes transparent stress shift in the context prior to mora-flip. In Harmonic Layer Theory, the mechanism depicted in (2) falls out from the novel mechanism of activity decay. Within a model based on activity, each cyclic optimization can result in a predictable activity loss for a certain phonological element. Structures that were optimized more often (=added at the stem layer) have hence the intrinsic potential of a different phonological behaviour compared to elements that are added later (=at the word layer).

(2) Kashaya in HLT: Late processes apply to a subset of the structure

a. Optimization ① SA, no IL or MF	b. Optimization ② IL and MF for previously optimized elements
Stem *IL *MF ✓SA	Stem Word ✓IL ✓MF — *IL *MF —

(SA=stress assignment, IL=iambic lengthening, MF=mora-flip)

This paper is structured as follows: In section 2, I present the relevant empirical facts about the metrical structure of Kashaya and the three processes of iambic lengthening, mora-flip, and stress assignment. Subsection 2.4 summarizes the ordering paradoxes that arise from the interaction of these processes in a classical stratal system where processes are assigned to a certain stratum. Section 3 gives a general introduction into Harmonic Layer Theory and its three main assumptions of 1) a cyclic optimization, 2) a single phonological grammar for each language, and 3) gradient activity for phonological elements. The main proposal of this paper is the Harmonic Layer Theory account of

Kashaya which is presented in section 4. Section 5 discusses alternative accounts of the Kashaya data. Since the Kashaya pattern and its challenges for any theory of the morpho-phonology interface has received considerable theoretical attention over the past four decades, this section considers multiple existing or imaginable alternative accounts: A Lexical Phonology account in 5.1, a parallel account based on constraint domains in 5.3, and two possible implementations within stratal OT, one based on two-step processes in 5.4 and one based on prosodic structures as process domains in 5.5. It is concluded that all of these alternatives face some serious theoretical or empirical problem that the present HLT account resolves. Since the analysis presented here differs in its empirical generalizations in one crucial respect from the generalizations derived by many alternative accounts, subsection 5.2 also motivates this decision with a detailed empirical discussion of final vowel shortening in Kashaya. I conclude in section 6.

2 Data: Metrical structure in Kashaya

Kashaya is a Pomoan language spoken in Sonoma County along the Pacific Coast of California and is the language of the Kashia Band of Pomo Indians of the Stewarts Point Rancheria. Its metrical structure and especially its relevance for the interface of phonology and morphology is discussed from a variety of perspectives and with different empirical foci in the work of Eugene Buckley (e.g. Buckley (1994*b*), Buckley (1994*a*), Buckley (1997), or Buckley (2017)). The data in Buckley’s work is based on his own fieldwork and the work by Robert Oswalt, in particular his dissertation Oswalt (1961). In addition to these published works, there is an excellent online resource for Kashaya which includes an online dictionary and sound files (Buckley et al., 2025). This paper cites most examples in the form given in Buckley (2017) where morpheme-by-morpheme-glosses are provided for each example. In the following, I focus on the interaction of three metrical

processes, namely iambic lengthening, the assignment of main stress, and mora-flip that changes initial CV:CV into CV.CV: . These three processes already illustrate a number of interesting challenges for the morpho-phonology interface but are only a subset of the processes restricted by morpho-syntactic factors in Kashaya which also include a number of segmental processes (cf. especially Buckley (1994*b*)). An exhaustive discussion of all facts is unfortunately beyond the scope of this paper. As was already alluded to above, the morphological structure of words is essential in understanding the metrical facts. The essential distinctions is the one into root- material and affix material, and the one into stem- and word-level affixes. Most of the following examples are verbs since those are morphologically more complex than nouns and hence show the relevant distinctions more easily. Verbal prefixes in Kashaya are instrumental prefixes and always part of the stem. All stem-level suffixes are optional and include, for example, those with adverbial meaning, aspect, or valence. Word-level suffixes express, for example, mood, evidentiality, or switch reference. In contrast to stem-level suffixes, one word-level suffix is obligatory.

2.1 Iambic Lengthening (=IL)

With this background, we can now turn to the phonological processes referring to metrical structure. As is illustrated in (3) and (4), there is predictable vowel lengthening in certain CV-syllables. Since it systematically applies to even-numbered syllables in a CV-sequence (3-a), it is best analysed as iambic lengthening (=IL) under the assumption that there are iterative, left-aligned feet. As (3-b) shows, CVC-syllables count as heavy and constitute a foot on their own and hence interrupt the symmetric IL for each even-numbered syllable. All examples in the following give the underlying form in the first line which includes square brackets denoting the stem- and word-boundaries and underlining for roots. The second line gives the phonological surface form.¹ To ease readability, IL is marked by $\overline{\text{V}}$. The surface forms include syllable boundaries and foot structure to facili-

tate linking the data to the generalizations. It goes without saying that this foot structure is already a theoretical assumption and part of the proposed account.²

(3) Iambic Lengthening

- | | |
|---|---|
| a. [[mo-mul-ɪf’-ed] -w]
(mo.m <u>ú:</u>)(li.tʃ’ <u>e:</u>)du
[[run-around-REFL-DUR] -ABS]
‘run in circles’ (B17:114) | b. [[kel-mul-ad-utʃed] -w]
(kéɫ)(mu.l <u>a:</u>)(du.tʃ <u>e:</u>)du
[[peer-around-DUR-DUR] -ABS]
‘keep peering around’ (B17:115) |
|---|---|

In contexts with either a prefix or a bisyllabic root, IL affects one syllable further to the right, namely the third and fifth CV-syllable in (4) and (5). This can be explained by initial syllable extrametricality that excludes any initial syllable if this syllable is not the only root-syllable, notated as "<>" below (cf. especially Buckley (1994a)).

(4) Initial syllable extrametricality and IL

- | | |
|--|---|
| a. [[tʃa-q ^h am-ala-w-ibiʃ] -w]
<tʃa>(q ^h a.m <u>á:</u>)(la.w <u>i:</u>)biʃ
[[by.knife-cut-down-MVMT-INCEP] -ABS]
‘start to cut downward’ (B17:115) | b. [[qaʔtʃ’aɫ-ad-utʃed] -w]
<qaʔ>(tʃa.t <u>á:</u>)(du.tʃ <u>e:</u>)du
[[cry-DUR-DUR] -ABS]
‘used to cry and cry’ (B17:115) |
|--|---|

The crucial restriction for our discussion of stratal paradoxes is the fact that IL never affects word-level suffixes, as can be seen in (5) where a CV-syllable surfaces as such in a foot-head. The example (5-a) is especially insightful since it involves only a CV-root that precedes a word-level suffix. The initial syllable of the word-level suffix must hence end up in the head-position of the only foot of the word and still remains short. This shows us that it cannot be the case that footing simply never re-applies beyond the stem-level. Such a re-analysis would wrongly predict *(sɪ)p^hi.la.

(5) IL never affects word-level suffixes

- | | |
|---|---|
| a. [[s’i] -p ^h ila]
(s’i.p ^h i)la
[[do] -DS.FUT]
‘if it happens’ (B17:120) | b. [[mo-matʃ-ed] -ela]
(mo.m <u>á:</u>)(tʃe.de)la
[[run-in-DUR] -PERF.IPFV]
‘I keep running in there’ (B17:120) |
|---|---|

IL hence shows a interstratal difference where stem- and word-level material behaves differently. IL is further restricted and never applies to final vowels, as can be seen in the examples in (6). They all consist of a trisyllabic root where the final syllable is expected

in the head position of the only foot of the word but the vowel remains nevertheless short.

(6) IL never affects word-final vowels

- | | |
|--|--|
| a. [[butaqa]]
<bu>(ta.qá)
'bear' (B94:175) | b. [[ʔot ^h ono]]
<ʔo>(t ^h o.nó)
'edible seaweed' (B94:128) |
|--|--|

Importantly, underlyingly long vowels remain long in final position as is shown in (7).

The contrast between monomorphemic (6) and (7) is essential for discussing the various analytical options to account for the absence of word-final IL (cf. especially 5.2).

(7) Underlyingly long root-vowels in word-final position

- | | |
|---|--|
| a. [[ʔit ^h a:]]
<ʔi>(t ^h á:)
spider
'spider' (B94:368) | b. [[ʔama:]]
<ʔa>(má:)
land
'land, place' (B94:368) |
|---|--|

2.2 Main stress assignment (=SA)

In all examples above, the final vowel of the first foot is marked with an acute accent. In Buckley (1994*b*) and subsequent work by Eugene Buckley, this is referred to as 'stress' whereas Oswalt (1961) describes it as a 'raised tone' which 'consists of a rise in pitch higher than that of the neighboring syllables' (Oswalt, 1961, 28). The formal analysis in section 4 represents this as a property of the head-syllable within the head-foot of a prosodic word and hence as classical main stress, often referred to as simple 'stress' below. Crucially, however, nothing hinges on this formal implementation and an alternative representation as, for example, a H-tone that docks to a head-vowel is as easily possible.

Stress assignment (=SA) puts stress on the initial iambic foot in all examples above. Interestingly, there are two exceptions to this leftmost main stress generalization. The first one is illustrated in (8), the second one is taken up later in 2.3. In all examples in (8), main stress is realized on the second foot-head and not on the leftmost foot-head. In (8-a+b), a bimoraic foot follows a monosyllabic initial one and stress hence surfaces on the third (non-extrametrical) syllable. In (8-c+d), a heavy syllable follows the initial

foot and stress hence surfaces on the second (non-extrametrical) syllable. The crucial difference between these examples and the ones above is that the leftmost foot in (8-a-d) is monosyllabic and contains an underlyingly long vowel. Note that the position of stress in (8-a-d) is further evidence that word-level material is indeed footed (even though it never undergoes IL) since stress predictably surfaces on foot-heads containing word-level material. The generalization is thus that main stress surfaces on the leftmost foot unless this is an initial foot containing an underlying long vowel.

(8) Second-foot main stress for underlyingly long initial vowels

- | | |
|---|---|
| <p>a. [[q'a:] -mela]
(q'a:)(me.lá)
[[leave.behind] -PERF.PFV
'I left (it) behind' (B17:121)</p> | <p>b. [[sima:q] -eti]
<si>(ma:)(qa.tí)
[[sleep] -CONC
'although he's asleep' (B17:121)</p> |
| <p>c. [[fula:m] -iʔba]
<fu>(la:)(máʔ)ba
[[be.sick] -COND]
'would get sick' (B17:117)</p> | <p>d. [[du-ʔya:q-ad-qa] -ba]
<duʔ>(ya:)(qán')(qa.ba)
[[by.finger-think-DUR-CAUS] -SS.PST]
'after thinking about it' (B17:116)</p> |

The avoidance of initial long stressed vowels can become opaque by vowel shortening which avoids superheavy CV:C-syllables in Kashaya. The long root vowel in (9-a)³ is hence shortened if a C-initial suffix follows. Interestingly, main stress surfaces on the second foot in this form, even though the initial vowel is shortened on the surface. This is contrasted in (9-b) with a form where the same root is followed by a V-initial suffix and the long vowel surfaces: Stress shift is transparent in (9-b) but opaque in (9-a).

(9) Main stress shift made opaque by vowel shortening

- | | |
|--|---|
| <p>a. [[siwi:tʃ] -meʔ]
<si>(wiʔʰ)(méʔ)
[[hang] -PL.IMP]
'hang (there)!' (pl) B94:241</p> | <p>b. [[siwi:tʃ] -in]
<si>(wi:)(tʃín)
[[hang] -ADV]
'while hanging' B94:241</p> |
|--|---|

This opacity can follow from assuming that vowel shortening to avoid superheavy syllables is a late process that does not apply at the point where SA applies (Buckley, 1992, 1994b). This is the solution that I adopt here as well (cf. 4.1.2). Vowel shortening,

however, is not central to any of the arguments that will follow.

Before turning to the final metrical process this paper focuses on, it should be emphasized that main stress in Kashaya is in fact a phrasal phenomenon. In each phrase, only a single main stress surfaces. Usually, it simply is the main stress of the initial word. In the following, I will largely ignore phrasal stress and focus on the opaque interaction of metrical processes at the word level. As is discussed in 4.1.2 and 7.1, the phrasal stress facts follows naturally from the current account if the default option is to realize only a single foot at the phrasal optimization.

2.3 Mora-flip (=MF)

The third metrical process interacting with both SA and IL is the so-called ‘mora-flip’ (=MF) which reverses the vowel length in a sequence of an initial (or post-extrametrical) long vowel followed by a short vowel in an open syllable: CV:CV. → CV.CV:, illustrated in (10) where the syllables exchanging their length are marked with CV.CV:.⁴ From a metrical perspective, MF creates the typologically least marked asymmetric iambic foot of a light syllable followed by a heavy syllable. In Buckley (1994*b*), this surface effect is derived by a process of ‘foot-flipping’ but this paper adopts the more neutral term ‘mora-flip’ which focuses on the autosegmental interpretation of the process as the de-association of one mora and the re-association of this mora to a neighbouring vowel.

(10) MF creates an asymmetric iamb

- | | |
|--|---|
| <p>a. [[t'e:t-ibitʃ] -ba]
 (te.ti:)(bíʃ')ba
 [[stand-up] -SS.PST]
 ‘having stood up’ (B17:117)</p> | <p>b. [[loq'o:tf-ad-uwad] -w]
 <lo>(q'o.tʃa:)(du.wá:du)
 [[move.noisily-along-DUR] -ABS]
 ‘make noise by moving around’ (B17:118)</p> |
|--|---|

In addition to the length-change, MF-context systematically show main stress on the second foot. This was already apparent in (10) and illustrating again with a minimal pair in (11). The two words in (11) are identical in terms of their segmental content and their segmental length distribution but they result from different underlying roots: (11-a) has an

underlying short vowel whereas (11-b) has a n underlyingly long vowel. The long surface vowel in the second syllable in (11-a) is a result of IL and is predictably stressed. The same long surface vowel in (11-b), on the other hand, results from MF and is not stressed.

(11) Minimal pair: MF initial foot is not stressed, IL initial foot is stressed

- | | |
|---|---|
| <p>a. [[tʃad-utʃed] -w]
 (tʃa.d<u>ú</u>)(tʃe.du)
 [[see-DUR] -ABS]
 ‘always sees’ (B94:183)</p> | <p>b. [[tʃa:d-utʃed] -w]
 (<u>tʃa.du</u>)(tʃe.dú)
 [[fly-DUR] -ABS]
 ‘always flies’ (B94:183)</p> |
|---|---|

The example in (12) proves that main stress is realized on the first foot in underlying CVCV:-sequences; that main stress is realized in the second and not the first foot in (10) and (11-b) is hence an exclusive property of MF-contexts. If one now recalls that MF-contexts involve an initial (or post-extrametrical) long vowel, there is an obvious parallel to the stress shift discussed in 2.2: Main stress avoids underlyingly long initial vowels and is realized on the second foot instead. If main stress assignment were sensitive to the underlying vowel length, the stress shift in (10) and (11-b) is completely transparent.

(12) Stress on underlying CV.CV:-

- [[ʃulot’o:-itʃ’-watʃ’] -w]
 <ʃo>(lo.t’ó:)(tʃ’i.waʔ)
 [[peel-REFL-DUR] -ABS]
 ‘keep peeling from oneself (pl)’ (B94:186)

A final restriction is that MF only applies to stem material and is hence blocked in (13-a) and (13-b) where the short vowel is part of a word-level affix. This restriction is absolutely parallel to IL which also only applies to stem material.

(13) No MF for word-level affixes

- | | |
|---|--|
| <p>a. [[q’a:] -mela]
 (q’a:)(me.lá)
 [[leave.behind] -PERF.PFV]
 ‘I left (it) behind’ (B17:121)</p> | <p>b. [[sima:q] -eti]
 <si>(ma:)(qa.tí)
 [[sleep] -CONC]
 ‘although he’s asleep’ (B17:121)</p> |
|---|--|

Even though word-level suffixes are never affected by MF, their shape matters for the application of the process. More concretely, MF is possible in example (14-a) because

the second syllable is open but impossible in (14-b) because the second syllable is closed. These examples involve the same stem combined with different word-level affixes. The relevant syllabification deciding whether MF is possible or not is only decided at the word-level since it depends on whether the following word-level suffix is V-initial (14-a) or C-initial (14-b). There is thus an ordering paradox which is very similar to the one identified for IL: MF only affects stem-level material but its context is only properly defined once all word-level material is present as well.

- (14) MF is fed/bled by word-level syllabification
- | | | | |
|----|---|----|---|
| a. | [[q'a:-tʃid] -w]
(q'a.tʃi:)dú
[[leave.behind-DUR] -ABS]
'keep leaving' (B17:121) | b. | [[q'a:-tʃid] -ba]
(q'a:)(tʃín')ba
[[leave.behind-DUR] -SS.PST]
'after leaving' (B17:122) |
|----|---|----|---|

2.4 Data summary: Ordering of metrical processes

The list in (15) summarizes the empirical generalizations about IL, SA, and MF under a stratal perspective where each process can apply at the stem, word, or phrase level.

- (15) Ordering account of Kashaya metrical processes
- | | | |
|-----|--|---------------|
| a. | Iambic Lengthening (IL) | |
| i. | only applies to stem material | Stem |
| ii. | but never to word-final vowels. | Word |
| b. | Mora-flip (MF) | |
| i. | only applies to stem material | Stem |
| ii. | but can be bled by syllabification of word-level material. | Word |
| c. | Stress assignment (SA) | |
| i. | avoids initial (or post-extrametrical) long vowels | |
| ii. | and can be counterbled by MF. | >MF |

An ordering paradox emerges for both IL and MF in (15). They only apply to stem material but their application contexts rely on information only present at the word-level.

For stress assignment, the challenge is different: Since MF changes initial long vowels into short ones and can hence counterbleed SA (15-c-ii), SA must apply before MF. Under

a standard model where the stem is the first optimization, this generalization is hence at odds with the generalization that MF should already apply at the stem-level (15-b-i). This ordering paradox is further illustrated for IL in (16) where the two options of IL as a stem-level (16-a) or word-level process (16-b) are considered. Crucially, IL is taken to only apply to non-final vowels to correctly exclude forms like *<bu>(ta.qa:), cf (6). It is shown that each option results in a misprediction for one of two crucial datapoints. If IL is a stem-level process (16-a), it underapplies for vowels which are stem-final but not word-final. And if it is a word-level process (16-b), it overapplies to word-level suffixes.

(16) IL: Ordering paradox

	[[mo-mul-ɪf'-ed] -u], cf. (3) (mo.m <u>ú:</u>)(li.tʃ <u>e:</u>)du	[[s'i] -p ^h ila], cf. (5) (si.p ^h <u>í</u>)la
a. Option A		
Stem stratum: IL	(mo.m <u>ú:</u>)(li.tʃed)	(sí)
Word stratum: –	*(mo.mú:)(li.tʃe)du ☹	(si.p ^h í)la ☹
b. Option B		
Stem stratum: –	(mo.mú)(li.tʃed)	(sí)
Word stratum: IL	(mo.m <u>ú:</u>)(li.tʃ <u>e:</u>)du ☹	*(si.p ^h <u>í:</u>)la ☹

The same is done for MF in (17) that interacts with closed syllable shortening (=CSS).

(17) MF: Ordering paradox

	[[q'a:-tʃid] -ba], cf. (14) (q'a:)(tʃín')ba	[[q'a:] -mela], (13) (q'a:)(me.lá)
a. Option A		
Stem stratum: MF	(<u>qa.tʃi:d</u>)	(q'a:)
Word stratum: CSS	*(qa.tʃín').bá ☹	(q'a:)(me.lá) ☹
b. Option B		
Stem stratum:	(qa:)(tʃíd)	(q'a:)
Word stratum: MF, CSS	(qa:)(tʃín').ba ☹	*(<u>q'a.me:</u>)lá ☹

Recall from (9) that there is good reason to assume that CSS is a late process applying after SA. I take it to be a word-level process in (17). Assigning MF to the stem-level wrongly results in overapplication for [[q'a:-tʃid] -ba] which in turn results in overapplication of CSS at the word level. The impossibility of super-heavy syllables hence wrongly results

in vowel shortening rather than the blocking of MF. And assigning MF to the word-level over-predicts it for word-level affixes as in [[q'a:] -mela].

I argue that these ordering paradoxes vanish in a cyclic model where processes can be restricted to elements which have been optimized previously. Whereas SA applies at the first optimization, IL and MF only apply to vowels that have been optimized before; i.e. to stem vowels during the word-level optimization. In the next sections, I show in detail how the simple generalization (17) falls out in a theoretical analysis based on Harmonic Layer Theory. Before proceeding, however, I want to emphasize that the empirical generalizations in (16) differ slightly from the generalizations derived in Buckley (1994*b*). Section 5.2 discusses these differences in detail and argues why the generalizations in (16) are more general and hence empirically adequate.

3 Theoretical background: Harmonic Layer Theory

Harmonic Layer Theory (=HLT Trommer, 2019; Zimmermann and Trommer, 2022; Zimmermann, 2025*b,c*; Tebay, 2023, 2025; Tebay et al., 2025) is a model of the phonology-morphology interface that shares its central architectural assumption with classical Lexical Phonology (Mohanan, 1982; Kiparsky, 1982, 1985) or stratal OT (Kiparsky, 2015; Bermúdez-Otero, 2018; Trommer, t.a.): The phonological evaluation is cyclic and derives increasingly bigger morpho-syntactic units. The application of phonological processes is hence interleaved with the concatenation of morphemes. The crucial difference between stratal models and HLT is the number of grammars that are assumed for each language. Whereas stratal accounts assume potentially different grammars that are active at different strata, HLT is based on a single grammar that cyclically optimizes for each language. That different strata can show different phonological behaviour falls out in HLT from the assumption that all linguistic symbols have a certain activity and that these ac-

tivities can gradiently differ (Gradient Symbolic Representations; =GSR, e.g. Smolensky and Goldrick, 2016; Rosen, 2016, 2019*a,b*). Previous work on GSR focused mainly on the explanatory power that gradient activity has for exceptionality (Zimmermann, 2018, 2019, 2021, 2025*a*; ?; Walker, 2019): If underlying representations of morphemes can differ in the activation of their phonological elements, different phonological behaviour can emerge for these elements. This prediction relies on the mechanism of gradient constraint violations. As a tendency, weak input elements are easier to delete or change than more active ones since their change violate faithfulness constraints to a lesser degree and weaker output elements are easier to tolerate in marked contexts than more active ones since they violate markedness constraints to a lesser degree. A cyclic model with GSR predicts the central mechanism of HLT: Predictable activity decay of elements. If the activity of certain phonological elements decays at each optimization, elements in earlier optimization cycles can be predicted to behave differently than comparable elements in later optimization cycles that were optimized more often. That different morpho-syntactic strata show divergent phonological behaviour is hence not a consequence from different grammars in HLT but from the cyclic optimization with the same grammar which is sensitive to the activity of elements. One main argument for this novel theory of the morpho-phonology interface is its restrictiveness compared to stratal alternatives. As is argued in Zimmermann (2025*c*), Tebay (2025), and Tebay et al. (2025), HLT inherently excludes *ABA-patterns of interstratal differences which are argued to be cross-linguistically unattested. Activity decay is not a built-in mechanism of HLT, it simply is a language-specific phonological repair. This is illustrated in tableau (19). A context-free markedness constraint *V (18-a) penalizes all vowels. Since its violations are evaluated gradiently and the constraint is hence sensitive to the activity of a vowel, a weaker vowel will violate it less severely than a more active vowel. Adjusting activity is now taken to be a possible phono-

logical repair which is penalized by specific faithfulness constraints like IDENTACT(V) for vowels (18-b). In addition to general IDENTACT(V) constraints, there are threshold constraints penalizing an activity adjustment that goes beyond a certain set amount. In our toy example (19), this amount is set to 0,2 by the constraint (18-c). The contrast between the constraints (18-d) and (18-e) illustrates a further assumption made here about the constraint inventory of HLT: Every markedness and faithfulness constraint exists in a gradient and a categorical version. The standard MAX constraint against deletion hence exists in a version that is violated depending on the input activity of the deleted element (18-d) and in a version that is insensitive to any activities (18-e). The latter constraints are given in double straight bars.

- (18)
- a. *V: Assign -x violation for every vowel V_x .
 - b. IDENTACT(V) (=IDENTACT(V) $\geq$.2): For every vowel V_y corresponding to input V_x , assign -(lx-yl) violation.
 - c. IDENTACT(V) $\geq$.2 (=IDENTACT(V) $\geq$.2): For every vowel V_y corresponding to input V_x , assign -(lx-yl) violation if (lx-yl) $>$ 0,2.
 - d. MAX(V): Assign -x violation for every input vowel V_x without a corresponding output vowel.
 - e. ||MAX(V)||: Assign -1 violation for every input vowel without a corresponding output vowel.

As all models based on gradient representation, the phonological optimization is based on Harmonic Grammar (Legendre et al., 1991, 2006; Pater, 2009; Potts et al., 2010). In such models, a harmony score ($\mathcal{H}$) is calculated for each candidate that adds all constraint violations times their respective constraint weights and the candidate with the harmony score closest to zero is optimal.

Tableau (19) illustrates the constraints (18) with an input that contains a fully active vowel /V₁/ and a weaker /V_{0,8}/. In the following, activity is expressed by subscripted numbers and 1 is taken to be the default underlying activity. Faithful realization of these vowels in candidate (19-a) induces -1,8 violations of the markedness constraint against vowels: The combined activity of all vowels in the output. Reducing the activity of both

vowels to 0,1 (19-b) reduces the violations of $*V$ to -0,2 but induces violations of both $\text{IDENTACT}(V)$ and $\text{IDENTACT}(V) \geq .2$. The latter is violated since the activity reduction for both vowels exceeds the threshold of 0,2. This violation is avoided in candidate (19-c) that only reduces the activity of both vowels by 0,2 and only violates the general $\text{IDENTACT}(V)$. This candidate emerges as the winner since it manages to avoid as many violations of $*V$ as possible without inducing a violation of high-weighted $\text{IDENTACT}(V) \geq .2$. Candidates (19-d) and (19-e) show that the complete deletion of a vowel is of course the ideal solution to avoid any violations of $*V$ but that this strategy is excluded in our toy grammar. The contrast between the violations of (19-d) and (19-e) illustrates the difference between categorical $\|\text{MAX}(V)\|$ and gradient $\text{MAX}(V)$: The latter will always favour deletion of a less active input element over deletion of a stronger input element.

(19) Activity decay for vowels

	$V_1 V_{0,8}$	$\text{IDA}(V) \geq .2$ 11	$\text{MAX}(V)$ 11	$\ \text{MAX}(V)\ $ 11	$*V$ 10	$\text{IDA}(V)$ 1	$\mathcal{H}$
a.	$V_1 V_{0,8}$				-1,8		-18,0
b.	$V_{0,1} V_{0,1}$	-1,6			-0,2	-1,6	-21,2
c.	$V_{0,8} V_{0,6}$				-1,4	-0,4	-14,4
d.	$V_{0,6}$		-1	-1	-0,6	-0,2	-28,2
e.	$V_{0,8}$		-1	-0,8	-0,8	-0,2	-28,0

In a system based on gradient representations, activity adjustment is hence an expected phonological process. And in a cyclic model where a grammar like (19) optimizes increasingly larger morphological domains, elements are predicted to gradually adjust their activity across layers, as shown in (20) for a cyclic optimization based on (19). A fully active vowel that enters the stem optimization will hence be optimized three times before it reaches the final phrase output and will lose 0,6 activity in total.

(20) Activity decay across layers, predicted from grammar (20)
Optimization

① Stem	$/V_1/ \rightarrow V_{0,8}$
② Word	$/V_{0,8}/ \rightarrow V_{0,6}$
③ Phrase	$/V_{0,6}/ \rightarrow V_{0,4}$

Predictable activity decay is the only possible source for different phonological behaviours across strata given that HLT is based on a single grammar for each language. Finally, we need to consider the options that the model predicts for epenthetic elements. GEN is free to insert elements of any activity but each option is penalized by a certain constraint, formulated in (21-b-d) for moras. The toy tableau (22) illustrates their effect with the optimization of an underlyingly mora-less vowel that needs to be dominated by an epenthetic mora in the output. Categorical (21-a) demanding that each vowel needs a mora, however, only excludes (22-a) where the vowel is mora-less but is equally happy with (22-b) where the mora has a minimal activity of -0,1, with (21-c) where the mora has the same output activity -0,8 as the vowel it dominates, and with (21-d) where it has the full activity of -1. The choice between these strategies termed ‘maximal epenthesis’, ‘minimal epenthesis’, and ‘dominance epenthesis’ is thus made by other constraints. Firstly, gradient DEP (21-b) will always prefer insertion of the weakest possible element, hence (22-a).⁵ The stronger the inserted mora is, the more fatal the DEP violations become until a fully active mora (21-c) violates it by -1. On the other hand, *WK constraints (21-c) are assumed to penalize any elements with an activity lower than 1 and hence show the reverse preference of preferring (22-b) over (22-c) over (22-d). Finally, there are constraints penalizing an element which dominates another element with a higher activity (21-d). It is hence violated by (22-d) but satisfied by (22-b) and (22-c) since the mora has at least the same activity as the vowel it dominates.

- (21) a. $\|HAVE_{\mu}\|$: Assign -1 violation for every vowel V that is not dominated by a mora.
b. *DEP(μ): Assign -x violations for every μ_x without an input correspondent.

- c. *WK(μ): Assign -1-x violations for every μ_x .
- d. $\|*WKD(\mu-V)\|$ ($=\|*WKD\|$): Assign -(x-y) violations for every V_x dominated by μ_y and $x>y$.

(22) Epenthesis (=Ep.) types: Exemplifying μ -tableau

	V_1	$\ HAVE[\mu]\ $	*DEP(μ)	*WK(μ)	$\ *WKD\ $
a. No Ep.	$V_{0.8}$	-1			
b. Maximal Ep.	$\begin{array}{c} \mu_1 \\ \\ V_{0.8} \end{array}$		-1		
c. Dominance Ep.	$\begin{array}{c} \mu_{0.8} \\ \\ V_{0.8} \end{array}$		-0,8	-0,2	
d. Minimal Ep.	$\begin{array}{c} \mu_{0.1} \\ \\ V_{0.8} \end{array}$		-0,1	-0,9	-0,7

As should be apparent from the counter-acting nature of the constraints, each of these epenthesis types can become optimal. Crucially, the constraint types in (21) are specified for different phonological elements and a language does not need to rely on a single epenthesis type. The Kashaya HLT account presented in the next section 4 in fact relies on a grammar that predicts dominance epenthesis for moras but maximal epenthesis for all other prosodic nodes.

4 A Harmonic Layer Theory account of Kashaya

Since HLT is based on a single grammar that cyclically optimizes, processes cannot directly be assigned to specific levels; their application at only certain levels is an epiphenomenon of activity decay. For Kashaya, I assume that both IL and MF apply only to vowels that already underwent one optimization and are hence weakened by 0,2; the processes do not apply to stronger vowels. Given the architectural assumption that the stem level is optimized on its own prior to the addition of word-level suffixes, this means that all stem vowels can undergo IL and MF at the word-level optimization. The ordering conundrums discussed above are resolved: Both IL and MF apply at a point where all

word-level material is present but they only apply to stem material as the subset of segments that were already optimized once. In contrast, SA applies at the first optimization and can hence be counterbled by MF. In the following subsections, the concrete constraint interactions predicting this analysis are discussed: IL is formalized in 4.2.1, MF in 4.2.2, and SA in 4.2.3. Before that, some central background assumptions about the activity decay in Kashaya 4.1.1 and the metrical structure 4.1.2 are made explicit. In Harmonic Grammar, every constraint violation is potentially relevant since even low-weighted constraints can gang up against higher-weighted ones. On the other hand, only the weighting relations between some crucial constraints are often relevant in a particular context. To resolve this presentational conundrum, I decided to focus on the crucial weighting relations in the following subsections and move full tableaux that include all relevant constraints and a more exhaustive list of candidates into an online appendix 7.2. The concrete constraint weightings assumed below are naturally only one possible weighting set that is compatible with the relevant weighting arguments. I manipulated the weights given below by hand to allow an easy illustration of the weighting arguments which implies that constraints in a relevant weighting relation have weights which are close together (i.e. the pairs 70 and 69 or 1.070 and 960).⁶

4.1 Background assumptions

4.1.1 Activity decay

The HLT account of the Kashaya ordering interactions of metrical processes is based on the predictable activity decay of three phonological elements: Vowels, moras, and feet. As can be seen in (23), the activity decay is assumed to be asymmetric and vowels and moras lose more activity at each optimization pass than feet. This is an expected property of any HLT system given that the existence and nature of activity decay emerges from the

interaction of a faithfulness and a markedness constraint referring to specific phonological elements (recall tableau (19)). Since no relevant gradient constraint refers to the activity of other phonological elements, they are assumed to retain their activity constant throughout all optimizations. It was shown above how activity decay is predicted from constraint interaction within HLT in tableau (19) and the specific decay in (23) is thus taken for granted in the following tableaux.

(23) Activity adjustments in Kashaya

	Optimization 1	Optimization 2	Optimization 3	...	
Feet	Ft_1	$Ft_{0,9}$	$Ft_{0,8}$	...	-0,1 Decay
Mora	μ_1	$\mu_{0,8}$	$\mu_{0,6}$	...	-0,2 Decay
Vowels	V_1	$V_{0,8}$	$V_{0,6}$	...	-0,2 Decay

The overview in (23) gives the activity decay in terms of the number of optimizations for each element. It will become apparent in the account below that vowels, moras, and feet can enter the derivation at different times. To mention the two most crucial contrasts: Vowels and moras on long vowels are underlyingly present and hence start with an input activity of 1 whereas feet and moras on short vowels are predictably assigned in the phonology. And secondly, all stem material is already present at the first optimization pass whereas word-level material is only added at the second optimization pass.

The final crucial background assumption is the activity of inserted elements. As was shown above in the discussion of tableau (22), there are three different possible epenthesis types in HLT, distinguished by a distinct constraint set for each phonological element. In Kashaya, feet undergo maximal epenthesis and hence enter the derivation with the default activity of 1 whereas moras undergo dominance epenthesis and hence enter the derivation with the same activity as the vowel they dominate. In the following tableaux, it is implied that moras on short vowels are underlyingly present. However, given that both vowels and moras decay by 0,2 and that moras undergo dominance epenthesis, the same output

is predicted if predictable moras on short vowels were underlyingly absent. There is hence no Richness of the Base issue relating to the (non)moraicity of short vowels and the account predicts the same outcome for underlyingly moraic or underlyingly mora-less short vowels.

The interesting opacities in Kashaya arise mainly at the interaction of two optimization cycles, namely the optimization of stem material ① and of word-material ②. A third optimization of phrase material ③ is only discussed in contexts where it is relevant. The account presented here hence shares its core architecture into three morpho-syntactic layers with classical Lexical Phonology or stratal accounts. This is, however, not inherent to HLT which only explores the predictions of a cyclic model based on activity decay.

4.1.2 Metrical structure

The analysis below focuses on the opaque interactions of IL, MF, and SA and proves how they fall out in HLT from predictably activity decay across layers. To allow the reader to focus on the crucial constraint interactions deriving IL, MF, and SA, a few general properties of the metrical structure are taken for granted in the following optimizations, listed in (24). All these properties are ensured by standard constraints on metrical structure (McCarthy and Prince, 1993; Kager, 1999) which are given in their full definitions and an illustrative weight for Kashaya in the Appendix 7.1.

- (24) General metrical properties of Kashaya
- a. Iterative left-aligned iambic feet are assigned
 - b. The initial σ is extrametrical unless this implies that no root material is footed
 - c. The left edge of the minimal prosodic word must be aligned with a foot
 - d. Trimoraic syllables are tolerated at the first optimization pass but will result in vowel shortening at the vowel's second optimization
 - e. Only the leftmost main stressed foot surfaces at the phrasal optimization

Assumption (24-a) is the most uncontroversial fact about Kashaya. Initial syllable extrametricality (24-b) is assumed to result in a recursive prosodic word structure (Ito and

Mester, 2009*a,b*) as in $\{\omega_{\text{Max}} \text{ bu } \{\omega_{\text{Min}} (\text{ta.qa})\}\}$ where a minimal prosodic word node must always be left-aligned with a foot (24-c) but a non-minimal prosodic word node can directly dominate a syllable at the left edge. Assumption (24-d) is responsible for the opaque interaction of SA and vowel shortening discussed above (cf. (9)). Vowels will remain long in closed syllables at their first optimization pass and only be shortened at their second optimization pass. At this point, however, SA already applied and vowel shortening can opaquely counterbleed SA. Since vowel shortening in superheavy syllables is not crucial for most of what follows, I will largely ignore it. The final assumption (24-e) derives the facts about the phrasal nature of main stress. Since the majority of crucial interactions between IL, MF, and SA apply at the first two optimizations, the phrasal layer will not be central in the following discussion.

4.2 Three metrical processes in Harmonic Layer Theory

4.2.1 Iambic lengthening

HLT resolves the ordering challenge between IL and word-finality by restricting IL to vowels that were weakened once. Stem vowels can hence undergo IL at their second optimization and thus at a point where word-level suffixes are present. A second effect from activity decay is that IL is only triggered by feet that are still strong enough. More concretely, feet are too weak to trigger IL at their third optimization pass. This means that word-level suffixes can never undergo IL at the phrasal level. The interaction of only two constraints predicts the basic IL pattern, namely the markedness constraint triggering IL, taken to be *EVENIAMB (25-a) demanding a strong foot-head⁷, and a faithfulness constraint aiming to block the application of IL, taken to be DEPAL(V- μ) (25-b) penalizing the insertion of a new association line between vowels and moras.⁸ Both constraints are crucially gradient and refer to the activity of feet (25-a) and vowels (25-b) respectively.

- (25) a. *EVENIAMB (= *EVIMB) W=69
Assign -x violations for every symmetric Ft_x that dominates two light syllables.
- b. DEPAL(V- μ) (=DEPAL) W=70
Assign -x violations for every input vowel V_x that is associated to a mora in the output but not in the input.

Their exemplifying weights of 69 and 70 already make clear that lengthening of a fully active input vowel will always be worse than a fully active light foot head: A -1 violation of DEPAL(V- μ) results in a worse harmony score than a -1 violation of *EVENIAMB. Any decay of vowels, however, will change this picture. Tableau (26) illustrates this with a context where two stem vowels undergo IL. The first optimization of only stem material is given in (26-①). Since the presence of iterative left-aligned iambic feet is taken for granted here (cf. 4.1.2), only two relevant candidates need to be considered, namely one without and one with IL. Candidate (26-①-a) is the former and simply parses all underlying segments into two bisyllabic feet. Since both foot-heads are only CV-syllables, *EVENIAMB is violated. Recall that the constraint is sensitive to the foot's activity, marked as a subscript to the brackets ' $()_x$ ' in the following. Since all feet are freshly inserted with a full activity of 1, two full -1 violations add up to a -2 of *EVENIAMB. Candidate (26-①-b) avoids these violations by lengthening the two vowels in foot head-position, hence violating DEPAL(V- μ). Since the constraint is relativized to the lengthened vowels' input activity, and both vowels are fully active in the input, the constraint is also violated by -2. Given that DEPAL(V- μ) has a higher weight than *EVENIAMB, there is no IL at the first optimization of stem material and (26-①-a) becomes optimal. Main stress is omitted for now; the optimal candidates are consequently only marked by (σ).

The output vowels in (26-①) show the predictable activity decay to 0,8 – and this is now the new input activity of those vowels for the second optimization (26-②). The word-level suffix /-u/ enters the derivation fresh from the lexicon and thus still has its full

activity 1 in the input. In the output, all vowel predictably decay, resulting in 0,6-activity for the stem vowels and 0,8-activity for the word-level suffix vowel. Crucially now, the decreased input activity of the stem vowels means that the lengthening candidate (26-②-b) only violates DEPAL(V- μ) by -1,6 given that two 0,8-vowels receive a new association to a mora. Since the feet's activity also decays to 0,9, the violations of *EVENIAMB in the non-lengthening candidate (26-②-b) are less serious than the ones for (26-①-b) and only add up to -1,8. Still, these violations are more fatal than the -1,6 violations of DEPAL(V- μ) and (26-②-b) emerges as optimal. IL of stem material hence becomes optimal at the second optimization. Even though both constraints become less fatal across layers since they refer to decaying elements, the faithfulness constraint refers to the faster-decaying element and is overridden at the second optimization cycle.⁹

(26) Stem-V's lengthen at ②: [[tʃa-q^ham-ala-w-ibitʃ] -ʔ], cf. (4)

	DEPAL 70	*EVIAMB 69	$\mathcal{H}$
① No IL			
(☞) a. <tʃa> (q ^h a _{0,8} .ma _{0,8}) ₁ (la _{0,8} .wi _{0,8}) ₁ (bi _{0,8} tʃ) ₁		-2	-138,0
b. <tʃa> (q ^h a _{0,8} .ma _{0,8}) ₁ (la _{0,8} .wi _{0,8}) ₁ (bi _{0,8} tʃ) ₁	-2		-140,0
② IL			
a. <tʃa> (q ^h a _{0,6} .ma _{0,6}) _{0,9} (la _{0,6} .wi _{0,6}) _{0,9} (bi _{0,6} ʔ) _{0,9}		-1,8	-124,2
(☞) b. <tʃa> (q ^h a _{0,6} .ma _{0,6}) _{0,9} (la _{0,6} .wi _{0,6}) _{0,9} (bi _{0,6} ʔ) _{0,9}	-1,6		-112,0

Tableau (27) now adds a context showing why word-level suffixes never undergo IL. It optimizes a trisyllabic stem, followed by a bisyllabic word-level suffix. As was already established in (26-①), the stem vowel in foot-head position remains short at the first optimization (27-①-a) since DEPAL(V- μ) has a higher weight than *EVENIAMB. At the second optimization, there are two CV-foot heads: The first one formed by a stem-level vowel with input activity 0,8 and the second one formed by a word-level suffix vowel with input activity 1. The two feet also have different activities. Since the first one was inserted at the first optimization, it is present in the input and hence decays to 0,9 in the

output. The second foot, however, is newly assigned and thus has the full output activity 1. Absolutely parallel to (26-②), the input vowel with 0,8 activity is predicted to undergo IL within a foot with activity 0,9. The activity distributions in the second foot, however, mirror the ones in (26-①) and an input vowel with activity 1 is predicted to remain short in a foot with activity 1. At the third (phrasal) optimization, the word-level affixes are in principle weak enough to undergo IL since they now only have 0,8 input activity. Since only a single foot surfaces at the phrasal optimization (cf. 4.1.2), however, it is now too late for the word-level suffix: It is not in foot-head position anymore in (27-③).

(27) Word-level vowels never lengthen I: [[mo-matʃ-ed] -ela], cf. (5)

	DEPAL 70	*EVIMB 69	$\mathcal{H}$
① No IL for St-vowel			
(13) a. (mo _{0,8} .ma _{0,8}) ₁ tʃe _{0,8} d		-1	-69,0
b. (mo _{0,8} .ma: _{0,8}) ₁ tʃe _{0,8} d	-1		-70,0
② IL for St- but not Wd-vowels			
a. (mo _{0,6} .ma _{0,6}) _{0,9} (tʃe _{0,6} .de _{0,8}) ₁ la _{0,8}		-1,9	-131,1
(13) b. (mo _{0,6} .ma: _{0,6}) _{0,9} (tʃe _{0,6} .de _{0,8}) ₁ la _{0,8}	-0,8	-1	-125,0
c. (mo _{0,6} .ma: _{0,6}) _{0,9} (tʃe _{0,6} .de: _{0,8}) ₁ la _{0,8}	-1,8		-126,0
③ Only a single foot and no IL for Wd-vowels			
(13) a. (mo _{0,4} .ma: _{0,4}) _{0,8} tʃe _{0,4} .de _{0,6} .la _{0,6}			-0,0
b. (mo _{0,4} .ma: _{0,4}) _{0,8} tʃe _{0,4} .de: _{0,6} .la _{0,6}	-0,6		-42,0

That only a single foot remains at the phrasal layer is hence the reason for the absence of IL at the third optimization in (27). With shorter stems, however, it is possible that a word-level suffix vowel ends up in the head position of the one surviving foot of the third optimization. That IL still does not apply in those contexts follows from the weakening of the foot, shown in (28) where a minimal CV-stem followed by a bimoraic word-level suffix is optimized. No IL is expected at the first optimization (28-①) since *EVENI-AMB only penalizes symmetric bisyllabic iambs and not monosyllabic iambs. This sub-minimal foot of course violates ||FTBIN|| but this violation is also not sufficient to trigger vowel lengthening since ||FTBIN|| only has a weight of 60 (cf. tableau (61) in the ap-

pendix). At the second optimization, the stem vowel is weak enough to undergo IL but it is now in the non-head position of the foot.¹⁰ The head of this foot is formed by a vowel belonging to a word-level suffix. Lengthening of this vowel (28-②-b) induces once again a full violation of DEP_{AL}(V-μ) since this vowel still had its full activity in the input. In addition, *EVENIAMB is only violated by -0,9 in (28-②-a) since the foot was already weakened once. The generalization that vowels never lengthen at their initial optimization hence holds for stem- and word-level vowels equally. The interesting expectation, however, is that a word-level suffix vowel should be weak enough to undergo IL at the next optimization, namely (28-③). And indeed, IL in (28-②-b) only induces a -0,8 violation of DEP_{AL}(V-μ) and thus the same weak violation that allowed IL for stem vowels in (27-②-b). The crucial difference, however, is the fact that the foot was now weakened twice already and only has a surface activity of -0,8 at its third optimization. And a -0,8 violation of *EVENIAMB is not enough to justify a -0,8 violation of DEP_{AL}(V-μ) – candidate (28-②-a) is hence optimal without any IL. That both vowels and feet decay at different rates thus predicts that the second optimization is the sweet spot for stem vowels: They are weak enough while the foot is still strong enough.

(28) Word-level vowels never lengthen II: [[s'i] -p^hila], cf. (5)

	DEPAL 70	*EVIAMB 69	$\mathcal{H}$
① No IL of St-vowel: Vowel too strong			
($\mathfrak{R}$) a. (s'i _{0,8}) ₁			-0,0
b. (s'i _{0,8}) ₁	-1		-70,0
② No IL of Wd-vowel: Vowel too strong			
($\mathfrak{R}$) a. (s'i _{0,6} .p ^h i _{0,8}) _{0,9} la _{0,8}		-0,9	-62,1
b. (s'i _{0,6} .p ^h i _{0,8}) _{0,9} la _{0,8}	-1		-70,0
③ No IL of Wd-vowel: Foot too weak			
($\mathfrak{R}$) a. (s'i _{0,4} .p ^h i _{0,6}) _{0,8} la _{0,6}		-0,8	-55,2
b. (s'i _{0,4} .p ^h i _{0,6}) _{0,8} la _{0,6}	-0,8		-56,0

Since IL is 'late' insofar as it only applies at the second optimization where word-level

affixes are already present, the ordering paradox discussed above is resolved. A categorical positional markedness constraint against final long vowels (29) (Prieto-Vives, 1994; Hayes, 1995; Myers and Hansen, 2006) will correctly block IL in word-final position but makes no misprediction about stem-final ones.

- (29) $||*V: \#||$ 10
Assign -1 violation for a final long vowel.

That $||*V: \#||$ decides the competition in contexts where IL is expected for a word-final vowel is shown in (30). A -0,9 violation of *EVENIAMB is tolerated at the second optimization of this word (30-②-a) since lengthening of the vowel induces not only a -0,8 violation of DEPAL(V- μ) but also a violation of $||*V: \#||$ (30-②-b). Since $||*V: \#||$ is still out-weighted by MAXAL(V- μ) (W=1.070) which protects underlying associations of vowels to moras, there is no misprediction that underlyingly long final vowels are shortened (cf. (7)); the constraint interaction thus only excludes newly derived long final vowels expected after IL. $||*V: \#||$ makes no misprediction for any of the tableaux shown above since all lengthened vowels in optimal candidates are in non-final position (but cf. the appendix for full tableaux for all contexts).

- (30) No IL for word-final vowels: [[butaqa]], cf. (6)

	DEPAL 70	*EVIAMB 69	$ *V: \# $ 10	$\mathcal{H}$
① No IL of St-vowel: Vowel too strong and final				
☞ a. $\langle bu_{0,8} \rangle (ta_{0,8}.qa_{0,8})_1$		-1		-69,0
b. $\langle bu_{0,8} \rangle (ta_{0,8}.qa_{0,8})_1$	-1		-1	-80,0
② No IL of St-vowel: Weak enough but not final				
☞ a. $\langle bu_{0,6} \rangle (ta_{0,6}.qa_{0,6})_{0,9}$		-0,9		-62,1
b. $\langle bu_{0,6} \rangle (ta_{0,6}.qa_{0,6})_{0,9}$	-0,8		-1	-66,0

The crucial weighting relations established for DEPAL(V- μ) and *EVENIAMB in this subsection are summarized in (31), together with the generalizations about IL that were proven in the tableaux above.

- (31) Crucial weighting relations: Iambic Lengthening
- a. No IL for stem material at ①: (26)-①-a, not (26)-①-b
 No IL for Wd material at ② (WdFt): (27)-②-b, not (27)-②-c
 $1 \times \text{DEPAL}(\text{V}_{-\mu}) > 1 \times * \text{EVENIAMB}$
 - b. IL for stem material at ②: (26)-②-b, not (26)-②-a
 $0,9 \times * \text{EVENIAMB} > 0,8 \times \text{DEPAL}(\text{V}_{-\mu})$
 - c. No IL for Wd material at ② (StFt): (28)-②-a, (28)-②-b
 $1 \times \text{DEPAL}(\text{V}_{-\mu}) > 0,9 \times * \text{EVENIAMB}$
 - d. No IL for Wd material at ③: (28)-③-a, not (28)-③-b
 $0,8 \times \text{DEPAL}(\text{V}_{-\mu}) > 0,8 \times * \text{EVENIAMB}$
 - e. No IL for final vowels: (30)-②-a, not (30)-②-b
 $1 \times \| * \text{V} : \# \| + 0,8 \times \text{DEPAL}(\text{V}_{-\mu}) > 0,9 \times * \text{EVENIAMB}$

4.2.2 Mora-flip

That word-level syllabification can bleed the apparent stem-level process of MF receives basically the same explanation as the one given for IL in 4.2.1: MF only applies to material that was weakened once and hence to stem material at its second optimization where word-level material is already present. Word-level syllabification can hence transparently bleed a process that only affects stem material.

The relevant constraint system predicting such an interaction is given in (32). The markedness constraint triggering MF is $\| * \text{INMONO} : * \#(\sigma) \|$ (32-a) penalizing initial monosyllabic feet. An initial long vowel is parsed into a single iambic foot and deassociating one mora of this vowel to a following short vowel will result in a more unmarked bisyllabic (CV.CV:)-foot.¹¹ Note that $\| * \text{INMONO} : * \#(\sigma) \|$ is categorical and hence insensitive to the activity of the foot. The faithfulness constraint counter-acting MF is $\text{MAXAL}(\text{V}_{-\mu})$ (32-b) penalizing any de-association of moras. It is crucially gradient which means that weakening of vowels will make MF possible; absolutely parallel to the gradient nature of $\text{DEPAL}(\text{V}_{-\mu})$ that allowed ‘late’ IL. The illustrative weights for $\| * \text{INMONO} : * \#(\sigma) \|$ and $\text{MAXAL}(\text{V}_{-\mu})$ are again very close to make the effect of gradient violations more appar-

ent: $\text{MAXAL}(\text{V}-\mu)$ has a higher weight than $\|*\text{INMONO}:\#\langle\sigma\rangle\|$ but a gradient violation of $\text{MAXAL}(\text{V}-\mu)$ will be less than a full violation of $\|*\text{INMONO}:\#\langle\sigma\rangle\|$. A constraint against weak domination $\|*\text{WKD}(\mu-\text{V})\|$ that was already introduced in 3, repeated for convenience in (32-c), will also become relevant. It excludes the re-association of a weak mora to a stronger vowel and will hence block MF between stem- and word-level vowels. Finally, an inviolable constraint $*\sigma^{\mu\mu\mu}$ excluding trimoraic syllables in Kashaya (cf. 4.1.2) is also shown in the tableaux below.

- (32)
- | | | |
|----|---|-------------------|
| a. | $\ *\text{INMONO}:\#\langle\sigma\rangle\ $ ($=\ *\text{IMO}:\#\langle\sigma\rangle\ $)
Assign -1 violation for every monosyllabic foot at the left edge of the prosodic word. | $\text{W}=960$ |
| b. | $\text{MAXAL}(\text{V}-\mu)$
Assign -x violations for every input vowel V_x that is associated to a mora in the input but not in the output. | $\text{W}=1.070$ |
| c. | $\ *\text{WKD}(\mu-\text{V})\ $
Assign -1 violation for every V_x dominated by μ_y and $x>y$. | $\text{W}=\infty$ |
| d. | $*\sigma^{\mu\mu\mu}$
Assign -1 violation for every syllable dominating more than two moras. | $\text{W}=960$ |

Tableaux (33) illustrate the effect of all these constraints in a context with successful MF. It optimizes a context with a prefix-less bisyllabic root; the initial root-syllable hence becomes extrametrical. Since the second syllable of the root contains an underlyingly long vowel, a violation of $\|*\text{INMONO}:\#\langle\sigma\rangle\|$ arises if all vowel length is realized faithfully (33-①-a). This violation can be avoided if one of the moras of the long vowel re-associates to the second vowel, creating an ideal asymmetric iamb (33-①-b). At the first optimization, however, this repair is too costly since it induces a -1 violation of both $\text{MAXAL}(\text{V}-\mu)$ and $\text{DEPAL}(\text{V}-\mu)$ and the sum of these violations is more fatal than the $\|*\text{INMONO}:\#\langle\sigma\rangle\|$ violation of candidates (33-①-a). Recall that the left edge of each minimal prosodic word is necessarily aligned with a foot (cf. 4.1.2) and non-parsing of the offending post-extrametrical syllable into a foot as another imaginable repair for $\|*\text{INMONO}:\#\langle\sigma\rangle\|$ is excluded. At the second optimization (33-②), the candidate with

faithful initial vowel length (33-②-a) still induces a -1 violation of categorical $\|*INMONO:*\#(\sigma)\|$. The MF candidate (33-②-b), however, only violates $MAXAL(V-\mu)$ and $DEPAL(V-\mu)$ by -0,8 each since these faithfulness constraints are sensitive to the decaying activity of the vowel. And these gradient violations are not as bad as the full violation of $\|*INMONO:*\#(\sigma)\|$: MF between stem vowels becomes optimal at the word-level. Note that both candidates in (33-②) show one instance of IL and hence an additional -0,8 violation of $DEPAL(V-\mu)$ each; completely predictable from the account in 4.2.1.

There are of course a number of other constraints that these candidates violate. Most notably, the non-MF candidate (33-①-a) induces a violation of $*EVENIAMB$ that the MF-candidate (33-①-b) avoids. On the other hand, sub-optimal (33-②-a) induces an additional -1 violation of $*EVENIAMB$ whereas optimal (33-②-b) induces an additional -1 $\|PARSE\|$ violation (cf. the full tableau (60) in the appendix). Violations of these constraints will never tamper with the crucial weighting relation between $\|*INMONO:*\#(\sigma)\|$ and $MAXAL(V-\mu)/DEPAL(V-\mu)$ since their weight is comparably low.

(33) [[loq'o:tf -ad -uwad] -u], cf. (10)

		$MAXAL(V-\mu)$	$*_{\sigma}^{\mu\mu\mu}$	$\ *IMo:*\#(\sigma)\ $	$DEPAL(V-\mu)$	
		1.070	960	960	70	$\mathcal{H}$
① No MF: Vowels are too strong						
(33) a.	$\langle lo_{0,8} \rangle (qo_{0,8})_1 (tfa_{0,8}.du_{0,8})_1 (wa_{0,8}d)_1$			-1		-960,0
b.	$\langle lo_{0,8} \rangle (qo_{0,8}.tfa_{0,8})_1 (du_{0,8}.wa_{0,8}d)_1$	-1			-1	-1.140,0
② Successful MF: Weakened vowels						
a.	$\langle lo_{0,6} \rangle (qo_{0,6})_{0,9} (tfa_{0,6}.du_{0,6})_{0,9} (wa_{0,6}.du_{0,8})_1$			-1	-0,8	-1016,0
(33) b.	$\langle lo_{0,6} \rangle (qo_{0,6}.tfa_{0,6})_{0,9} (du_{0,6}.wa_{0,6})_{0,9} du_{0,8}$	-0,8			-1,8	-912,0

Since MF can only apply at the word-level optimization, it will be trivially affected by the syllabification of word-level material and the look-ahead conundrum discussed above is resolved. This is shown in the optimizations (34) that derive context (14) where word-

level syllabification bleeds MF. Although MF at the second optimization (34-②-b) only induces -0,8 violations of both DEP_{AL}(V-μ) and MAX_{AL}(V-μ), it crucially also induces a violation of categorical $\sigma^{\mu\mu\mu}$ since an illicit trimoraic syllable results if the mora ‘flips’ into a syllable that is closed by a coda after the syllabification of word-level suffixes. The violation of $\|*INMONO:*\#(\sigma)\|$ in optimal (34-②-a) is hence tolerated in such a context. An option not considered in the discussion here is the complete deletion of a mora, hence simple shortening ($*(qa_{0,6}.tʃi_{0,6}d)_1$ for (34-②)). This option is excluded by the high weight of MAX-_{μINI} penalizing the deletion of a mora associated to the first vowel in a foot.¹²

(34) [[qɑ:tʃid] -ba], cf. (14)

	$\ *WkD\ $	MAX _{AL} (V-μ)	$\sigma^{\mu\mu\mu}$	$\ *IMO:*\#(\sigma)\ $	DEP _{AL} (V-μ)	
	∞	1.070	960	960	70	$\mathcal{H}$
① No MF						
☞ a. (qɑ _{0,8} ː) ₁ (tʃi _{0,8} d) ₁				-1		-960,0
b. (qɑ _{0,8} .tʃi _{0,8} :d) ₁		-1	-1		-1	-2.100,0
② No MF: Weak vowels but wrong syllabification						
☞ a. (qɑː _{0,6}) _{0,9} (tʃi _{0,6} d) _{0,9} ba _{0,8}				-1		-960,0
b. (qɑ _{0,6} .tʃiː _{0,6} d) _{0,9} (ba _{0,8}) _{0,9}		-0,8	-1		-0,8	-1.872,0

The previous tableaux established that – as IL – MF is a ‘late’ process only de-associating moras of vowels that were already optimized once and can hence transparently be fed or bled by the syllabification of material added at the second optimization. Why MF can never affect word-level suffixes is shown in (35) where the stem contains only a CV:-sequence. At the second optimization (35-②), the addition of a word-level suffix provides a potential host for a de-associated mora (35-②-b). And we know that de-association only causes -0,8 MAX_{AL}(V-μ) violation at this point. That association to a vowel of a word-level suffix induces a full violation of DEP_{AL}(V-μ) is in itself not a good enough reason

to block MF in (35-②-b) since DEPAL(V- μ) has a considerably lower weight than $\|*IN-MONO:*\#(\sigma)\|$. However, associating a stem-level mora to a word-level vowel induces a different fatal problem, namely a mora which dominates a vowel with more activity, violating $\|*WKD(\mu-V)\|$. The non-MF candidate (35-②-a) hence becomes optimal. Activity decay to moras and the dispreference for weak domination thus ultimately sets up a domain for MF: It can only apply between vowels that underwent the same number of optimization cycles. Since the activity of moras is crucial for the potential violations of $\|*WKD(\mu-V)\|$, I included an abbreviated autosegmental structure in (35).

(35) [[q'a:] -mela], cf. (8)

	$\ *WKD\ $	MAXAL(V- μ)	$*\sigma_{\mu\mu}$	$\ *IMO:*\#(\sigma)\ $	DEPAL(V- μ)	
	∞	1.070	960	960	70	$\mathcal{H}$
① No MF: Vowel too strong						
a. $\begin{array}{c} \mu_{0.8} \quad \mu_{0.8} \\ \diagdown \quad \diagup \\ (q \ a_{0.8})_1 \end{array}$				-1		-960,0
b. $\begin{array}{c} \mu_{0.8} \\ \\ (q \ a_{0.8})_1 \end{array}$		-1				-1.070,0
② No MF: St- μ too weak for Wd- μ						
a. $\begin{array}{c} \mu_{0.6} \quad \mu_{0.6} \quad \mu_{0.8} \quad \mu_{0.8} \\ \diagdown \quad \diagup \quad \quad \\ (q \ a_{0.8})_{0,9} (m \ e_{0.8} \ l \ a_{0.8})_1 \end{array}$				-1		-960,0
b. $\begin{array}{c} \mu_{0.6} \quad \mu_{0.6} \quad \mu_{0.8} \quad \mu_{0.8} \\ \quad \diagdown \quad \diagup \quad \\ (q \ a_{0.8} \ m \ e_{0.8})_{0,9} (l \ a_{0.8})_1 \end{array}$	-1	-0,8			-1	$-\infty$

MF is hence a ‘late’ process since moras can only de-associate from vowels that were optimized once. That it never affects word-level material then follows since a stem-mora is too weak to associate to a stronger word-level vowel. The crucial weighting relations predicting these state of affairs are summarized in (36).

(36) Crucial weighting relations: MF

a. No MF at ①: (34)-①-a, not (34)-①-b

$$1 \times \text{DEPAL} + 1 \times \text{MAXAL}(\text{V}-\mu) > 1 \times \|\text{*IMO:*\#}(\sigma)\|$$

b. MF at ② between stem-V's: (34)-②-b, not (34)-②-a

$$1 \times \|\text{*IMO:*\#}(\sigma)\| > 0,8 \times \text{MAXAL}(\text{V}-\mu) + 0,8 \times \text{DEPAL}$$

c. No MF at ② between stem- and Wd-level V's: (35)-②-a, not (35)-②-b

$$1 \times \|\text{*WKD}(\mu-\text{V})\| + 0,8 \times \text{MAXAL}(\text{V}-\mu) + 1 \times \text{DEPAL} > 1 \times \|\text{*IMO:*\#}(\sigma)\|$$

4.2.3 Main stress assignment

This final subsection now adds the derivation of main stress to the analysis. Recall that the main challenge discussed in 2.4 for main stress assignment was the different treatment of initial underlying CV.CV:-sequences and those that derive from MF. Whereas main stress is realized on the leftmost foot in the former context, it is realized one foot further to the right in the latter context. It is shown in this subsection that this falls out from the cyclic account provided here where MF is a process that only applies at the second optimization cycle. At the first optimization where main stress is assigned, a MF context hence still has an initial post-extrametrical long vowel CV:_. And those syllables are in fact the ones that cause predictable stress realization on the second foot anyway. Main stress in MF contexts can hence be reduced to the exceptional avoidance of (CV:)-feet that is systematic in Kashaya, even though this context might be destroyed on the surface by later vowel length manipulations.

The formal implementation of this intuition is based on the constraints in (37). Firstly, $\|\text{ALIGN}(\text{HD};\text{L})\|$ (37-a) prefers main stress to be on the leftmost foot. The one context where this default preference is avoided are initial/post-extrametrical long vowels where leftmost main stress induces a violation of the markedness constraint $\|\text{*INL-GHD:*\#}(\acute{\text{V}})\|$ (37-b). It is reminiscent of $\|\text{*INMONO:*\#}(\sigma)\|$ which triggers MF (32-a). Both constraints conspire in constraining the initial foot in Kashaya: It ideally contains

no long vowels and if those are unavoidable, then the long vowel should at least not be stressed. There is thus a general positional markedness preference for unmarked binary (CV.CV(:/C))-feet at the beginning of words (cf., for example, Zoll (1998); Crosswhite (2001); Walker (2005, 2011) for the concept of positional markedness). In addition, the faithfulness constraint $\|IDENT_{FT}(HD)\|$ (37-c) is crucial in the account which will ensure that the main stress usually remains in the foot that was optimal at its initial assignment. This ensures the opacity of MF contexts: Even though the marked #CV:-configuration is repaired later, main stress will still not retract back to the unmarked initial foot.

- (37)
- a. $\|ALIGN(HD;L)\|$ W=10
Assign -n violations for every Ft that intervenes between the head-foot and the left edge of the prosodic word.
 - b. $\|*INLGH D:*\#(\acute{V}):\|$ ($=\|*ILH:*\#(\acute{V}):\|$) W=151
Assign -1 violation for every main-stressed vowel in the initial syllable of a prosodic word.
 - c. $\|IDENT_{FT}(HD)\|$ ($\|ID_{FT}(HD)\|$) W=150
Assign -1 violation for every input head-foot which is not a head-foot in the output.

Before turning to concrete tableaux, recall from 4.1.2 that each word in Kashaya minimally contains one foot. The non-assignment of foot structure is hence no valid strategy to avoid a violation of $\|*INLGH D:*\#(\acute{V}):\|$. The first tableaux illustrate the generalization that main stress is leftmost unless this implies a stressed long initial vowel. Tableau (38) completes the optimization (26) with the newly introduced SA constraints. As was established in (26), there is no IL at the first optimization. As (38-①) shows, main stress is predictably assigned to the leftmost foot at this first optimization pass. Later IL does not change anything about the unmarkedness of this configuration and main stress remains.

(38) Main stress on the leftmost foot: [tʃa-q^ham-ala-w-ibitʃ] -ʔ], contd.

	ILH:($\acute{V}$):	IDFT(HD)	ALIGN(HD;L)	$\mathcal{H}$
	151	150	10	
① Leftmost main stress but no IL				
☞ a. <tʃa> (q ^h a _{0,8} .má _{0,8}) ₁ (la _{0,8} .wi _{0,8}) ₁ (bi _{0,8} tʃ) ₁				-0,0
b. <tʃa> (q ^h a _{0,8} .ma _{0,8}) ₁ (la _{0,8} .wí _{0,8}) ₁ (bi _{0,8} tʃ) ₁			-1	-10,0
② IL				
☞ a. <tʃa> (q ^h a _{0,6} .má _{0,6}) _{0,9} (la _{0,6} .wi _{0,6}) _{0,9} (bi _{0,6} ʔ) _{0,9}				-35,0
b. <tʃa> (q ^h a _{0,6} .ma _{0,6}) _{0,9} (la _{0,6} .wí _{0,6}) _{0,9} (bi _{0,6} ʔ) _{0,9}		-1	-1	-195,0

This is contrasted with a post-extrametrical long vowel in (39). Leftmost main stress (39-①-a) hence induces a fatal violation of ||*INLGH:*($\acute{V}$):||. Realization of main stress one foot to the right (39-①-b) avoids this violation and only induces a -1 violation of lower-weighted ||ALIGN(HD;L)||. At the second optimization cycle, this ||ALIGN(HD;L)|| violation is still tolerated. Shifting the main stress to the initial foot (39-②-b) induces a violation of ||*INLGH:*($\acute{V}$):|| and also a violation of ||IDENTFT(HD)|| given that the main stressed input foot would lose its head-status.

(39) Avoidance of H on long vowels: [du-ʔya:q-ad-qa] -ba], cf. (8)

	ILH:($\acute{V}$):	IDFT(HD)	ALIGN(HD;L)	$\mathcal{H}$
	151	150	10	
① Stress on the second foot				
a. <du _{0,8} >(yá _{0,8}) ₁ (qa _{0,8} d) ₁ qa _{0,8}	-1			-151,0
☞ b. <du _{0,8} >(ya _{0,8}) ₁ (qá _{0,8} d) ₁ qa _{0,8}			-1	-10,0
② Stress remains				
☞ a. <du _{0,6} >(ya _{0,6}) _{0,9} (qá _{0,6} d) _{0,9} (qa _{0,6} .ba _{0,8}) ₁			-1	-10,0
b. <du _{0,6} >(yá _{0,6}) _{0,9} (qa _{0,6} d) _{0,9} (qa _{0,6} .ba _{0,8}) ₁	-1	-1		-301,0

Tableau (40) adds the opaque stress assignment in MF context to the optimization (34).

For SA, the initial optimization (40-①) is very similar to the one in (39-①): To avoid a stressed initial long vowel (40-①-a), stress is assigned to the second foot even though this implies a violation of $\|ALIGN(HD;L)\|$ in optimal (40-①-b). At the second optimization (40-②), MF now applies and removes the offending initial long vowel. Main stress can now be on the initial foot (40-②-b) to avoid a violation of $\|ALIGN(HD;L)\|$ without causing an additional violation of $\|*INLGH\|$. However, this induces a fatal violation of $\|IDENT(HT)\|$ since a foot that was the head in the input is not the head-foot in the output anymore. An unmarked main stress pattern that was assigned at the first optimization can hence not be changed in a later step of the derivation.¹³ It is important to recall that $\|IDENT(HT)\|$ only cares about the head-status of a foot, not about the head-status of a syllable. The main stress foot that dominated /tʃa.du/ in optimal (40-①-b) dominates different material /du.wa/ in optimal (40-②-a). Nevertheless, it is still the same (second) foot that never changed its status of being a foot-head.

(40) Opaque stress shift in MF-contexts: [[loq'o:tʃ -ad -uwad] -u], contd.

	$\ *ILH:*\#(\acute{V})\ $	$\ IDENT(HT)\ $	$\ ALIGN(HD;L)\ $	$\mathcal{H}$
	151	150	10	
① Stress on the second foot: Initial long vowel				
a. $\langle lo_{0,8} \rangle (qo_{0,8})_1 (tʃa_{0,8}.du_{0,8})_1 (wa_{0,8}d)_1$	-1			-151,0
b. $\langle lo_{0,8} \rangle (qo_{0,8})_1 (tʃa_{0,8}.dú_{0,8})_1 (wa_{0,8}d)_1$			-1	-10,0
② MF but stress remains: Head faithfulness and opacity				
a. $\langle lo_{0,6} \rangle (qo_{0,6}.tʃa_{0,6})_{0,9} (du_{0,6}.wá_{0,6})_1 du_{0,8}$			-1	-10,0
b. $\langle lo_{0,6} \rangle (qo_{0,6}.tʃá_{0,6})_{0,9} (du_{0,6}.wa_{0,6})_1 du_{0,8}$		-1		-150,0

The opacity of SA and MF thus arises from the application of operations at different optimization cycles and the fact that MF only applies at the second optimization, in contrast to early SA. The relevant weighting relations are summarized in (41).

- (41) Crucial weighting relations: SA
- a. Leftmost main stress avoids initial long vowels: (39)-①-b, not (39)-①-a
 $1 \times ||*ILH:*\#(\acute{V}:)|| > 1 \times ||ALIGN(HD;L)||$
 - b. Main stress remains on the second foot at ② in MF contexts:
(40)-②-a, not (40)-②-b
 $1 \times ||IDFT(HD)|| > 1 \times ||ALIGN(HD;L)||$

5 Alternative accounts

5.1 The Lexical Phonology account in Buckley (1994b)

In Buckley (1994b) and Buckley (1994a), a Lexical Phonology account of Kashaya is presented. Importantly, IL, SA, and MF are only a small subset of what is derived in this detailed analysis which covers a plethora of segmental and length-related processes. Future has to prove that the HLT account developed here can be extended to the complete set of interstratal differences discussed meticulously in Buckley (1994b). Five lexical levels are assumed for the verbal suffixes of Kashaya, followed by one post-lexical level. As is argued in Buckley (2017), the five lexical levels can ‘with minimal changes’ be reduced to only a stem (=level 1-3) and a word-level (=level 4-5) for suffixes (p.113). I follow this two-way distinction here. IL is assumed to be a stem-level process. No ordering conundrum between IL and word-level information arises since the absence of word-final IL is explained by a final vowel shortening rule which applies at the word stratum. This solution is discussed in more detail in the following subsection 5.2.

The ordering conundrum between MF and word-level affixation identified in 2.4 is resolved by assuming that MF is a two-step process. At the stem level, an initial (or post-extrametrical) long vowel followed by a short vowel creates an ‘anti-iamb’, a marked structure where the initial heavy syllable forms a foot which appends the following CV syllable to its right by a process of ‘CV-Adjunction’ (Buckley, 1994b, 195-197). At the word-level, a process of ‘foot-flipping’ applies that changes this marked structure into an

unmarked (CV.CV:)-foot (Buckley, 1994*b*, 200-209). Foot-flipping of course only applies if word-level syllabification still leaves the appended syllable light. That MF never applies to word-level suffixes straightforwardly follows since the anti-iamb-creation only applies at the stem-level and foot-flipping only applies to anti-iamb contexts. The look-ahead problem between MF and word-level syllabification hence receives an interesting representational solution: Potential candidates for MF are marked as such at the stem-level but the process is only completed later. Stress shift is formalized as foot-extrametricity: Initial feet starting with a long vowel are extrametrical. Crucially, the context for foot extrametricality includes an optional appended syllable and hence an anti-iamb structure. Buckley (1997) criticizes that in this account, ‘[f]oot extrametricality must be formulated with reference to the dubious generalization ‘begins with Cvv’’ (p.27). In essence, however, the solution to stress shift in MF contexts proposed in Buckley (1994*b*) and in the present proposal is very similarly based on an intermediate representation: MF-contexts are indistinguishable from initial (or post-extrametrical) long vowels at the point where SA applies given that both start with a long vowel.

After the advent of Optimality Theory, Eugene Buckley puts forth several arguments against such an account (e.g. Buckley, 1995, 1996, 1997). One main argument is conceptual and emphasizes that it remains a coincidence under such an account that word-level affixes behave differently for both IL and MF, a fact which is evaluated as especially undesirable since both processes conspire to form the perfect iambic foot (Buckley, 1997, 34). In Buckley (1995) and Buckley (1996), this argument is extended and it is argued that a single constraint preserving vowel length is able to derive the fact that word-level suffixes are exempt from IL, MF, and final vowel shortening (cf. 5.3). Another potentially critical point about the proposed account is the assumption of cumulative extrametricality that it crucially relies on. As is discussed in detail in Buckley (1994*a*), cumulative

extrametricality in the Kashaya account is still restricted by peripherality since elements on different tiers – syllables and feet – are extrametrical and hence still peripheral at their respective tier. The extrametricality of feet is proposed by Hayes (1995) but the empirical adequacy of this concept has been questioned in, for example, McCarthy (2003). This argument is taken up in Buckley (2017) where it is argued that ‘it would also be possible to treat it as a shift in the accent onto the following foot at a late stage of the derivation’. I’m not sure, however, what such a solution would look like. That MF contexts also show non-initial stress only follows if SA is taken care of as early as possible and hence at a point where MF has not yet applied (my proposal) or is not yet complete (the proposal in Buckley (1994*b*)). If stress is shifted late in the derivation, a MF context should be indistinguishable from an underlying CV.CV:-context where stress is predictably on the first foot (cf. example (12)). The next subsection discusses another problem with the account in Buckley (1994*b*) and Buckley (1994*a*), namely the fact that a final vowel lengthening process is assumed to explain the absence of IL on word-final vowels. It is concluded that the broader empirical generalization discussed in 2.1 which extends beyond verbal contexts cannot fall out from an account based on such a vowel shortening rule.

5.2 Against the final vowel shortening rule in Buckley (1994*b*)

This subsection takes a closer look at the explanation put forth in Buckley (1994*b*) for the absence of word-final IL, namely the existence of a final vowel shortening process. To that end, I want to discuss such a process in a broader empirical context. My conclusion is that there is no definition of final vowel shortening that would explain all the absences of word-final IL without overgeneralizing it to contexts where word-final long vowels indeed surface. In Buckley (1994*b*), the ordering paradox for IL and word-finality is discussed in a footnote as the paradox that ‘no vowel which ends up in word-final position can undergo Iambic Lengthening, but still the rule must apply to vowels which are stem-final

at the time when the rule is active' (Buckley, 1994*b*, 194). It is resolved by assuming a final vowel shortening process. This implies a Duke-of-York derivation where a stem-final vowel might undergo IL at the stem-level but is shortened again if it is still final at the word-level. This vowel shortening rule is taken to be obligatory for verbs and optional for nouns (Buckley, 1994*b*, 312). In addition, word-final shortening is restricted to morphologically derived contexts (Buckley, 1994*b*, 373). The evidence for this process in verbal contexts comes from contexts where the imperative suffix /-i/ is added to stems ending in long vowel and a short vowel surfaces (42-a). This context involves regular hiatus-avoiding vowel deletion which (with only some exceptional contexts) results in the surfacing of the first vowel, exactly as in (42). That this vowel loses its length, however, is indeed surprising compared to similar word-internal contexts where the length of the surfacing first vowel is faithfully preserved (cf. (12) for an example). And the evidence for noun-final vowel shortening comes from compounding where either member of the compound can optionally shorten its vowel (43). This is one of the arguments in Buckley (1994*b*) to treat compounding as the combination of two independent prosodic words.

- (42) Final vowel shortening in derived contexts: Hiatus-avoiding vowel deletion
- | | | | | |
|----|---|----|--|-----------|
| a. | [[hi-s'a:] -i]
his'a
[[with.the.body-break] -IMP]
'break!' | b. | [[q'a:] -i]
q'a
[[leave] -IMP]
'leave him/her!' | (B94:312) |
|----|---|----|--|-----------|
- (43) Final vowel shortening in derived contexts: Compounds
- | | | | | |
|----|--|----|---|-----------|
| a. | [[ʔatʃaʔ]] [[ʔi ^h ya:]]
ʔatʃaʔ ʔi ^h ya
[[man]] [[bone/strong]]
'strong man' | b. | [[ʔi ^h yu:]] [[ʔima:ta]]
ʔi ^h yu ʔima:ta
[[ice]] [[woman]]
'ice woman' | (B94:371) |
|----|--|----|---|-----------|

There are at least three arguments against the assumption that the absence of final IL is due to final vowel shortening. Firstly and most importantly, it is not sufficient in explaining all contexts where IL does not apply to word-final vowels. Subsection 2.1 contrasted underived monomorphemic words and showed that a) they can end in an underlyingly

long vowel (cf. [ʔitʰa:] ‘spider’ (7)), and that b) the final vowel does not undergo IL (cf. [butaqa] ‘bear’ (6)). These facts cannot fall out from a final vowel shortening process restricted to derived contexts since they are monomorphemic forms. Another fact that makes the formulation of a general final vowel shortening process at least more complex is the fact that there are exceptions to it, even in morphologically derived environments. For one, morphological lengthening can derive final long vowels. As Buckley (1998a) points out, nouns marking a place can be converted into an adverbial marking movement along or toward that place by final vowel lengthening, shown in (44-a+b). In addition, final vowels resulting from MF can also surface finally. This is shown in (44-c) where /-e:/ precedes a long vowel resulting from MF. This suffix is analysed as a lexical clitic which imposes a word-boundary on the preceding stem and is parsed into a prosodic phrase itself (Buckley, 1994b, 301) which explains why it surfaces as long in all contexts. However, if /e:/ is not a word-final long vowel in (44-c), then the preceding /a:/ is.

- (44) Final long vowels
- | | | | | |
|----|--------|--------------------|----|---------------------------------|
| a. | ʔahʃa | ‘house’ | c. | [[miku:t -ad] -e:] [[mu]] |
| | ʔatʃa: | ‘to or at a house’ | | <mi>(<u>ku.ta:</u>)(dé:) (mú) |
| b. | dono | ‘mountain, hill’ | | [[hum-DUR] -NONFIN] [[3]] |
| | dono: | ‘uphill’ (B98:187) | | ‘he keeps humming’ (B94:186) |

The data discussed above shows that only word-final long vowels expected in certain contexts (result of IL, compounding, hiatus-avoiding V-deletion) are systematically avoided; others (underlying, result of morphological lengthening, result of MF) are perfectly licit. It is unclear to me how a rule-based Lexical Phonology account as the one in Buckley (1994b) could capture such a generalization. A final vowel shortening rule can be ordered before MF or morphological lengthening to predict long final vowels in those contexts but this still leaves the existence of underlying final long vowels unexplained.

For completeness, it needs to be mentioned that an alternative explanation for the absence of word-final IL is put forth in Buckley (1997). There, it is argued that it is simply

an epiphenomenon that falls out from the fact that word-level suffixes never participate in IL and that ‘[d]ue to the morphology of the language, every verb ends in a suffix which is not-lengthening’ (p.11). However, this argument only extends to verbs and thus the forms that most of the previous work focuses on. The generality of the absence of word-final IL that also extends to non-verbal context (cf. (6)) hence remains unexplained.

The OT-implementation presented here, on the other hand, can predict the facts in (44) since different markedness and/or faithfulness constraints can be either higher- or lower-weighted than $||*V:\#||$ penalizing long final vowels. It was already discussed above (cf. especially tableau (30)) that underlying final long vowels are preserved since $MAXAL(V-\mu)$ ($W=1.070$) out-weighs $||*V:\#||$ ($W=10$) but those expected from IL are avoided since a violation of $||*V:\#||$ and -0,8 violations of $DEPAL(V-\mu)$ ($W=70$) are more than a -0,9 violation of $*EVENIAMB$ ($=69$). That long vowels derived by MF remain long in final position is also already predicted by the account presented here: An additional $||*V:\#||$ violation ($W=10$) in a successful MF-candidate (cf., for example, (59)-c) will never cause this candidate to be sub-optimal compared to one not undergoing MF since the MF-causing $||*INMONO:*\#(\sigma)||$ is much more important ($W=960$).

5.3 A parallel OT account with domain-specific constraints

In Buckley (1995), Buckley (1996) and Buckley (2017), a parallel OT account for the Kashaya metrical facts is discussed. One main argument for such an OT-implementation is the fact that it predicts the conspiracy that word-level suffixes are exempt from IL, MF, and final vowel shortening from a single faithfulness constraint whereas it remains a coincidence under a rule-based account as the one in Buckley (1994*b*) or Buckley (1994*a*). In these proposals, the different behaviour of stem- and word-level affixes follows from the concept of ‘constraint domains’ which assumes that constraints can be restricted to only certain domains of a morphologically complex form. In spirit, this assumption is hence

very similar to the lexical indexation of constraints to certain morphemes (Pater, 2006, 2010; Finley, 2009) but has the advantage that it captures the fact that all morphemes sensitive to a specialized constraint must be contiguous; a fact which remains coincidental under constraint indexation (Buckley, 2017). A faithfulness constraint like Q-IDENT which demands preservation of vowel length is assumed to be particularized to a domain mirroring the stem and to another domain including word-level affixes – and the version sensitive to the latter is high-ranked, excluding both IL and MF for word-level affixes. The look-ahead problem discussed for MF and also for non-final IL vanishes in such a completely parallel account: All material is evaluated at once and word-level suffixes are hence present at the point where IL and MF apply.

By nature, such a completely parallel account is inherently unable to predict opacity. One instance of opacity in Kashaya is the stress-shift in MF and closed syllable shortening contexts where an originally long but short surface vowel remains unstressed. In the parallel OT account presented in Buckley (1997), the stress shift for initial long vowels is predicted from a general SKIPFOOT. The opaque stress shifting contexts are then argued to have in common that there is an alternation: The morpheme surfaces with a long initial vowel in some (non-shortening, non-MF) contexts where stress shift is predictable but as short in other contexts. Stress shift is then derived by OO-faithfulness constraint demanding identical stress patterns in paradigmatically related forms (Kenstowicz, 1995; Benua, 1997). Although the account is parallel, it indirectly allows reference to morphological information by employing paradigmatic faithfulness. And this explanation fails to derive stress shift from the same underlying principle, despite its obvious relation to long vowels in all contexts. Two completely orthogonal explanations – SKIPFOOT and OO-faithfulness for stress – predicts stress shift for initial long surface vowels, those shortened by closed syllable shortening, and MF-contexts. In contrast, the present proposal can de-

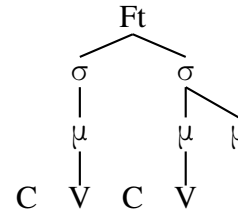
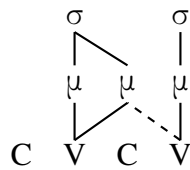
rive stress shift in all three contexts from the same general mechanism (=the constraint $||*INLGHD:*\#(\acute{V}:)||$) since there are intermediate forms in this cyclic architecture.

5.4 A stratal OT account

The most obvious translation of an account based on ordered rules into OT is a stratal OT account where potentially different constraint rankings optimize increasingly bigger morpho-syntactic units (Kiparsky, 2015; Bermúdez-Otero, 2018; Trommer, t.a.). One option to resolve the ordering paradox identified in 2.4 within stratal OT is the representational solution that is also employed in the Lexical Phonology account in Buckley (1994*b*) and (Buckley, 1994*a*): Both IL and MF are decomposed into two-step processes. They are hence initiated at the stem level where they affect all viable stem material but can only be finalized at the word level where they might be bled by the new contexts created by the suffixes added at this stratum. In Buckley (2017), a two-step decomposition of MF is discussed which is based on an intermediate structure like (45-a): An initial/post-extrametrical CV:CV-sequence results in a structure where the second mora of the initial vowel is doubly-linked. It hence starts the flipping to the following vowel but can't complete it. Only the word-level phonology allows to completely associate the doubly-linked mora to a new vowel. And this association is then of course blocked if a the second vowel is followed by a coda. Such a representational account can be extended to IL as well if one assumes that all symmetric CV-CV-feet cause insertion of an additional mora for the second syllable at the stem level to make this foot-head heavy but that this mora cannot associate to a vowel yet and remains 'hanging' (Trommer and Zimmermann, 2014) as in (45-b). Only at the next optimization within the word-stratum will this mora associate to its vowel – and only if this does not result in a superheavy syllable or a final long vowel.

(45) IL and MF as a two-step processes

- a. First step of MF: Doubly linked mora b. First step of IL: Hanging mora



Buckley (2017) criticizes such representational accounts as essentially diacritic: A marked intermediate representation which is otherwise unmotivated in the language is necessary to resolve the look-ahead problem. And even if one accepts the existence of language-internally unmotivated intermediate structures as a natural consequence of any account based on co-grammars, there is a specific issue with the intermediate representations necessary for Kashaya. The doubly-linked mora structure in (44-a) should be harmonically bounded given any standard set of constraints on prosodic structure. Even if GEN can generate a doubly-linked mora, it necessarily violates some markedness constraint. On the other hand, a doubly-linked mora does not improve on any imaginable constraint formulation that requires heavy foot heads. Either such a constraint is the one against symmetrical feet used here (cf. (25)) or it is a more general one against light foot-heads. If the former is assumed, the structure in (44-a) still violates such a constraint given that the foot now contains two heavy syllables. And if the latter is assumed, the constraint is not violated in (44-a) but it also was not violated in the competing structure without the double-linking since this contained a monosyllabic heavy foot. It is hence not clear to me which grammar based on independently motivated constraints can successfully predict the necessary intermediate structure (44-a) in a stratal OT alternative.

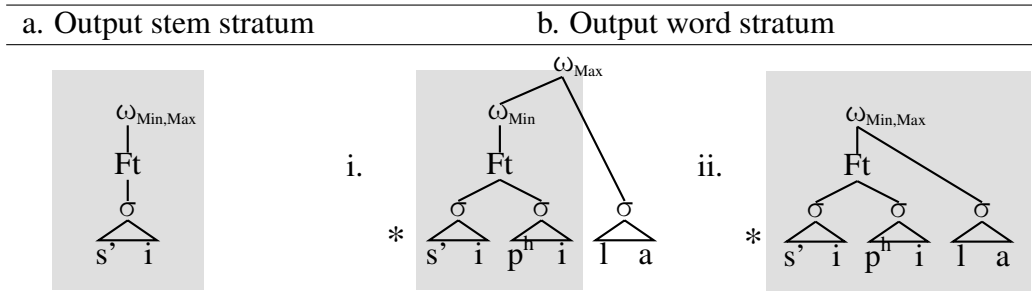
5.5 Prosodic domains in stratal OT

The HLT account presented here resolves the ordering paradoxes discussed in 2.4 by the intuition that processes can apply to only a subset of the material evaluated. This mech-

anism can also be modeled as a prosodic restriction, a classical solution in cyclic models where ‘prosodic structure assigned in an earlier cycle can persist, and continue to affect the application of phonological processes’ (Bermúdez-Otero, 2011, 2028). An example for such a stratal account where prosodic structure differences mirror earlier cyclic optimizations is presented in Bermúdez-Otero and Luis (2011) for pronominal enclitics in European Portuguese which are incorporated under a recursive maximal prosodic word node, in contrast to other word-level suffixes that are part of a minimal prosodic word. Transferred to Kashaya, this would mean that MF and IL can be re-analysed as processes that apply at the word-level but only to vowels within a minimal prosodic word which is co-extensive with the material optimized at the stem stratum. Since we know that word-level suffixes are iteratively footed (recall examples like $[[\text{si}ma:q] -eti] \rightarrow <si>(ma:)(qa.tí)$, cf. (8)) and that stem-level suffixes can be footed together with word-level syllables (recall examples like $[[\text{mo}-matf-ed] -ela] \rightarrow (mo.má:)(tfe.de)la$, cf. (5)), it is impossible to simply assume that stem-level affiliation is faithfully transferred as membership within the minimal prosodic word. Rather, the minimal prosodic word in Kashaya can only dominate syllables that it already dominated in its first optimization but it can lose syllables which are newly integrated into the maximal prosodic word formed at the word stratum (e.g. $\{\omega_{Max} \{\omega_{Min} (mo.ma)\} (tfe.de)la\}$). However, the assumptions discussed so far run into problems for super-short stems like (46). At the stem stratum, a degenerate monosyllabic foot parsed into a prosodic word results (46-a). At the word stratum, the addition of the suffix allows the creation of an unmarked bisyllabic foot (46-b), apparent from the surface stress on the second syllable. There is, however, no integration of this foot under prosodic word structure which is consistent with the generalization that IL only applies within minimal prosodic words (=grey in (46)) but is still absent in (46-b). Irrespective of whether two recursive prosodic word nodes are present (46-b-i) or only a single one (46-b-ii),

the initial foot is necessarily part of the minimal prosodic word, simply because every minimal prosodic word needs to dominate something and a maximal prosodic word is impossible without the simultaneous presence of a minimal prosodic word. Both imaginable prosodic structures in (46-b) thus mispredict overapplication of IL for the head-vowel under the generalization that IL only applies within the minimal prosodic word.

(46) Strata and prosodification: $[[s'i] - p^hila] \rightarrow (s'i.p^hila)$, cf. (5)



That only the subset of stem-material is affected by a process can hence not fall out from prosodic domains alone. In contrast, it follows in HLT since the generalization that processes are restricted to weakened material does not impose a minimality restriction on the size of this domain, in contrast to the prosodic alternative (46).

6 Conclusion

This paper proposed a novel analysis of Kashaya and the various ordering challenges that the interaction of metrical processes poses for any theory of the morpho-phonology interface. It was argued that a HLT system where activity of phonology elements decays predictably across cyclic optimizations with the same grammar correctly predict that processes can be restricted to a morpho-syntactic sub-domain while still being potentially counterbled or -fed by material which is part of a bigger morpho-syntactic domain. Within HLT, this follows from the generalization that IL and MF only affect vowels that were already optimized once. Both processes thus apply only after the addition of word-level suffixes but only to stem material. This cyclic model also offers an immediate expla-

nation for the opaque stress shift in MF contexts. Since the former applies at the first optimization, its context can be destroyed by the latter at the next optimization. Although this intuition is seemingly identical to the one that is predicted from a stratal system where processes are confined to prosodic domains, the Kashaya case study showed that the predictions in fact differ and Kashaya can follow under a HLT model but not one based on prosodic domains. The main argument in this paper is hence an argument for HLT based on its ability to correctly account for the challenging interaction of metrical processes in Kashaya. The existing arguments for HLT focus on its restrictiveness (Zimmermann, 2025c; Tebay, 2025; Tebay et al., 2025): Since the model is based on a single phonological grammar that optimizes cyclically within one language, the realm of possible interstratal differences within one language is smaller than the one predicted from a stratal system based on co-grammars. The Kashaya case study in this paper adds the argument that the model is not only more restrictive it indeed makes qualitatively different predictions about the possible interaction of processes across morpho-syntactic domains. HLT is still a novel framework that proposes a radically different view of the morpho-phonology interface. Case studies like these help us to understand its predictions and potential better. On the other hand, future work has to show whether the model is indeed empirically adequate for all the challenges posed by interstratal differences and process interactions across morpho-syntactic domains. Even within Kashaya, this paper can only provide a first important step: It focuses on the interaction of metrical processes but had to leave the interstratal differences at the segmental level for future work.

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Notes

¹Since only Buckley (2017) provides morpheme-by-morpheme glosses, all other glosses are added by myself based on the affix list provided in Buckley (1994b:pp.389-392) and Buckley et al. (2025). The abbreviations follow the Leipzig glossing rules (Haspelmath et al., 2008). Additional abbreviations are: CONC-concessive, DS-different subject, INCEP-inceptive, MVMT-movement, PERF-performative, SS-same subject. I follow the notational practice used in Buckley (2017) to notate glottalization as <Cʔ> and also unified some notations to IPA that are notated differently in some of the sources: <š> is replaced by <ʃ>, <t> is used for an apico-alveolar stop, <: > for vowel length. In addition, I follow Buckley (2017) in using <ɬ> instead of <c>. A single underlying form for each morpheme is given instead of allomorphs. This is especially apparent for the pervasive absolutive suffix that predictably surfaces as [w] after a vowel, as [u] after /d/ and as [ʔ] after consonants other than /d/. I follow the argumentation in Buckley (1994b) that the underlying form must be /-w/ (cf. especially p.61) and notate it as /-w/ throughout. I also avoid the abstract underlying /i/ allomorph that is always changed into [u] after /d/ (Buckley, 1994b, 115, FN 28). The sources are cited as follows: Buckley (1994b)='B94', Buckley (1998a)='B98', and Buckley (2017)='B17'.

²The foot structure given here is different from the one in Buckley's work in some contexts since it dispenses with initial foot extrametricality.

³Note that I glossed /meʔ/ as single plural imperative morpheme which simplifies the fact that /ʔ/ behaves as a separable element and is glossed as a clitic on its own in the sources.

⁴The long vowel of a MF configuration can also arise in hiatus contexts (e.g. [[si-de-afʔ'-iyifʔ'] -in] → <si>(de.ɬ'i):(yi.ɬ'ín) 'while they were sailing along' (Buckley, 2007, 118)). Since vowel hiatus always results in a long vowel in initial or post-extrametrical position, this structure is not interestingly different for the account presented here and I will abstract away from this difference in the following.

⁵That this is the 'weakest' possible output element implies an additional threshold against weaker elements since GSR in principle allows infinite weakness ($\mu_{0,0001}$, $\mu_{0,00001}$,...). Tableau (47) relies on an inviolable constraint against output elements with an activity below 0,1. Since minimal epenthesis is not relevant in Kashaya, I won't delve further into this discussion.

⁶I hence decided against using a learning tool like MaxEnt (Hayes, 2009) since the exemplary weights calculated with such a program are not as insightfully paired. It goes without saying that the harmony scores of the full tableaux in 7.2 were checked with a calculator program ("numbers") and not calculated by hand.

⁷It is crucially only restricted to branching iambs and hence does not penalize sub-minimal CV-feet, absolutely parallel to ASYM Buckley (1995, 1996, 1997).

⁸Recall from the discussion in 4.1.1 that it is assumed here that short vowels are underlyingly provided with a mora in order to not add to many non-contrastive violations of DEP_{AL}(V- μ). However, the account would work equally well if short vowels are underlyingly mora-less.

⁹Note that the following tableaux only list the contrastive DEP_{AL}(V- μ) violations and omit any violations that predictably arise if an underlyingly short and thus potentially mora-less vowel is provided with the mora necessary for its prosodic integration. If all four underlyingly short vowels in (47-①) are underlyingly mora-less, -4 additional DEP_{AL}(V- μ) violations arise for each candidate (cf. the discussion in 4.1.2).

¹⁰The alternative candidate $*(s'i_{0,6})(p^h i_{0,8-0,9} la_{0,8})$ with a lengthened initial stem vowel in a foot on its own is excluded by a constraint introduced later, namely $\|*INMONO:*\#(\sigma)\|$ penalizing initial long vowels. Cf. the appendix (61) for a full tableau of context (28) which includes this candidate.

¹¹Initial CVC.-syllables will also violate $\|*INMONO:*\#(\sigma)\|$ but this violation is unavoidable given that consonant deletion or mora-less codas are penalized by higher-weighted constraints.

¹²The high weight of this constraint is also decisive in hiatus-avoiding vowel deletion contexts. As is discussed in detail in Buckley (1994b) or Buckley (1998b), vowel deletion results in a short vowel for all but the initial syllable. I take this to follow from the following weights: MAX- μ_{INI} =860, MAX- μ =230, and

*V:=165. In non-initial contexts, the need to preserve a mora is not enough to warrant the creation a new long vowel (=violation of *V:) and the insertion of a new association (=violation of DEPAL(V-μ)) whereas both these violations are tolerated in initial position.

¹³The only context where the head-status of a foot is re-assigned in the second optimization are #(<CV>)CV:(C)-stems that enforce marked main stress on the initial long vowel at the first optimization. At the second optimization, this marked stress is repaired via stress shift if possible. Such a context is optimized in the appendix in (59).

7 (Online) Appendix

7.1 Deriving the metrical background facts

The following list of constraints with their exemplifying weight shows how the facts about the Kashaya metrical system that were taken for granted in the analysis above (cf. (24)) are derived. It repeats some of the constraints that were already included in the account above to show their relevant interaction with additional constraints.

Iterative left-aligned iambic feet are assigned

- (47) a. $\|RHT:I\|$ ∞
Assign -1 violation for every non-head that is final in a foot.
- b. $\|ALIGN(PRWD,L;FT,L)_\sigma\|$ 40
Assign -1 violation for every syllable that intervenes between the left edge of a prosodic word and the left edge of a prosodic foot.

$\|ALIGN(PRWD,L;FT,L)_\sigma\|$ has a lower weight than $\|PARSE\|$ to allow iterative footing.

Degenerate feet are avoided unless they are the only chance to avoid a head-foot containing only a long vowel

- (48) a. $\|PARSE\|$ W=35
Assign -1 violation for every syllable which is not integrated into a foot.
- b. $\|FTBIN\|$ W=60
Assign -1 violation for every foot which only dominates a single mora.

The initial σ is extrametrical unless this implies that no root material is footed

- (49) a. $\|RTINFT\|$ ∞
Assign -1 violation for every word where no root material is dominated by a foot.
- b. $\|*\#\sigma INFT\|$ 5.000
Assign -1 violation for every maximal prosodic word which is left-aligned with a foot.

In Buckley (1997), the relevant constraints are FT-ROOT ‘The root is dominated by a foot’ for (49-a) and $ALIGN(FT,L;SYL,R)$ ‘Align the left edge of every foot with the right edge of a syllable’ for (49-b). One might argue that (49-a) can only have an effect at the first optimization and that an equivalent of Bracket Erasure excludes any reference to root material at later optimizations. That initial syllable extrametricality persists into

later optimizations can then follow from assuming that a maximal prosodic word cannot be deleted again (and cannot be co-extensive with another minimal prosodic word).

The left edge of the minimal prosodic word must be aligned with a foot

- (50) $\| \text{ALIGN}(\text{PRWD}_{\text{MIN}}, \text{L}; \text{FT}, \text{L}) \|$ 10.000
Assign -1 violation for every left edge of a minimal prosodic word that is not aligned with the left edge of a foot.

Every word needs minimally one left-aligned foot. $\| \text{ALIGN}(\text{PRWD}_{\text{MIN}}, \text{L}; \text{FT}, \text{L}) \|$ is out-weighted by $\text{ALIGN}(\text{FT}, \text{L}; \text{PHRASE}, \text{L})_{\text{FT}}$ (52-a) and hence violated at the phrase layer where only one foot surfaces.

Trimoraic syllables are excluded and repaired via vowel shortening

- (51) a. $\| \text{WEIGHTBYPOSITION} \|$ ∞
Assign -1 violation for every coda consonant that is not moraic.
- b. $\text{MAXAL}(\text{V}-\mu)$ $\text{W}=1.070$
Assign -x violations for every input vowel V_x that is linked to a μ that has no output correspondent.
- c. $*\sigma^{\mu\mu\mu}$ $\text{W}=960$
Assign -1 violation for every syllable dominating more than two moras.

Weakening of vowels allows closed syllable shortening only at the second optimization pass of a vowel: $1 \times \text{MAXAL}(\text{V}-\mu) > 1 \times * \sigma^{\mu\mu\mu}$ but $1 \times * \sigma^{\mu\mu\mu} > 0,8 \times \text{MAXAL}(\text{V}-\mu)$. CV:C-syllables are hence tolerated at the stem-layer optimization; resulting in the opacity discussed in (9).

Phrasal stress follows, I assume since only a single foot surfaces at the third (=phrasal) optimization where words are combined. This assumption differs from the assumptions made in Buckley's work where iterative feet are still present at the phrasal layer. The noniterativity of phrasal footing follows from (52-a), a constraint that is irrelevant for the first two optimizations where no prosodic phrase is present. Only the pressure to avoid a violation of $\| * \text{INLGHD}: * \#(\acute{\text{V}}) \|$ causes the surfacing of two left-aligned feet at the third optimization, phrasal stress hence still avoids initial long vowels. Constraint (52-b) ensures that it is a main-stressed foot that surfaces. It ensures that the opacity of stress shift in MF contexts is retained at this late optimization: If the main-stressed foot is the second one of the initial word, phrasal stress will remain there. This implies that an additional inviolable constraint penalizes any deletion of association lines between feet and syllables, feet can hence only gain new syllables; they can never lose syllables that they originally dominated. That there is never IL for stem-level vowels that end up in

a new head-position at the third optimization follows from (52-c). Note that (52-c) can also be violated in successful MF candidates like (33) – the relative low weight of this constraint makes sure that it doesn't block MF to satisfy $\|*INMONO:*\#(\sigma)\|$ (W=960), it only blocks application of phrasal IL to satisfy $*EVENIAMB$ (W=69).

- (52) a. $\|ALIGN(FT,L;PHRASE,L)_{\sigma}\|$ 71
 Assign -1 violation for every syllable that intervenes between the left edge of a prosodic phrase and the left edge of a prosodic foot.
- b. $\|MAXHD\|$ 70
 Assign -1 violation for every input head-foot without an output correspondent
- c. $DEPAL(V_{\mu})_{HD}$ 14
 Assign -x violations for every input head vowel V_x that is associated to a mora in the output but not in the input.

7.2 Full tableaux

(53) [[tʃa-q^ham-ala-w-ibitʃ] -ʔ]

	∞ *WkD	MAXAL 1.070	$\sigma^{*}\mu\mu\mu$ 960	*IMo:*(σ) 960	*ILH:*(V:) 151	*V:## 10	IdFt(HD) 140	DEPAL 70	*EVI _{MB} 69	FtBIn 60	P _{PARSE} 35	ALIGN(HD;L) 10	$\mathcal{H}$
① Leftmost main stress but no IL													
☞ a. <tʃa> (q ^h a _{0,8} .má _{0,8}) ₁ (la _{0,8} .wi _{0,8}) ₁ (bi _{0,8} tʃ) ₁									-2				-138,0
b. <tʃa> (q ^h a _{0,8} .má: _{0,8}) ₁ (la _{0,8} .wi: _{0,8}) ₁ (bi _{0,8} tʃ) ₁								-2					-140,0
c. <tʃa> (q ^h a _{0,8} .má _{0,8}) ₁ la _{0,8} .wi _{0,8} (bi _{0,8} tʃ) ₁									-1		-2		-139,0
d. <tʃa> (q ^h a _{0,8} .má: _{0,8}) ₁ (la _{0,8} .wi _{0,8}) ₁ (bi _{0,8} tʃ) ₁								-1	-1				-139,0
e. <tʃa> (q ^h a _{0,8} .ma _{0,8}) ₁ (la _{0,8} .wí _{0,8}) ₁ (bi _{0,8} tʃ) ₁									-2			-1	-148,0
② IL													
a. <tʃa> (q ^h a _{0,6} .má _{0,6}) _{0,9} (la _{0,6} .wi _{0,6}) _{0,9} (bi _{0,6} ʔ) _{0,9}									-1,8				124,2
☞ b. <tʃa> (q ^h a _{0,6} .má: _{0,6}) _{0,9} (la _{0,6} .wi: _{0,6}) _{0,9} (bi _{0,6} ʔ) _{0,9}								-1,6					112,0
c. <tʃa> (q ^h a _{0,6} .má: _{0,6}) _{0,9} la _{0,6} .wi _{0,6} (bi _{0,6} ʔ) _{0,9}								-0,8			-2		126,0

(54) [[mo-matʃ-ed] -ela]

	∞ *W _K D	1.070 MAXAL	960 $\sigma^{\mu\mu\mu}$	960 *IM _O :*#(σ)	151 *IL _H :*#($\acute{V}$;)	10 *V:##	140 IdFt(H _D)	70 DEPAL	69 *EvIMB	60 FTBIN	35 PARSE	10 ALIGN(H _D ;L)	$\mathcal{H}$
① No IL													
☞ a. (mo _{0,8} .má _{0,8}) ₁ (tʃe _{0,8} d) ₁									-1				-69,0
b. (mo _{0,8} .má _{0,8}) ₁ (tʃe _{0,8} d) ₁								-1					-70,0
② IL for stem- but not word material													
a. (mo _{0,6} .má _{0,6}) _{0,9} (tʃe _{0,6} .de _{0,8}) ₁ la _{0,8}									-1,9		-1		-166,1
☞ b. (mo _{0,6} .má _{0,6}) _{0,9} (tʃe _{0,6} .de _{0,8}) ₁ la _{0,8}								-0,8	-1		-1		-160,0
c. (mo _{0,6} .má _{0,6}) _{0,9} (tʃe _{0,6} .de _{0,8}) ₁ la _{0,8}								-1,8			-1		-161,0
d. (mo _{0,6} .má _{0,6}) _{0,9} tʃe _{0,6} .de _{0,8} .la _{0,8}								-0,8			-3		-161,0
③ A single foot remains and no IL for word material													
☞ a. (mo _{0,4} .má _{0,4}) _{0,8} tʃe _{0,4} .de _{0,6} .la _{0,6}											-3		-105,0
b. (mo _{0,4} .má _{0,4}) _{0,8} tʃe _{0,4} .de _{0,6} .la _{0,6}								-0,8			-3		-161,0

(55) [[butaqa]]

	$8 \parallel *W_K D \parallel$	1.070	960	960	151	10	140	70	69	60	35	10	
	$\parallel *W_K D \parallel$	$MAXAL$	$\sigma^{\mu\mu\mu}$	$\parallel *IM_O : * \#(\sigma) \parallel$	$\parallel *ILH : * \#(\acute{V}) \parallel$	$\parallel *V : \# \parallel$	$\parallel IdFt(HD) \parallel$	$DEPAL$	$*EvIMB$	$\parallel FTBIN \parallel$	$\parallel PARSE \parallel$	$\parallel ALIGN(HD;L) \parallel$	$\mathcal{H}$
① No IL													
☞ a. <bu _{0,8} > (ta _{0,8} .qa _{0,8}) ₁									-1				-69,0
b. <bu _{0,8} > (ta _{0,8} .qa _{0,8}) ₁						-1		-1					-80,0
② No IL for final V													
a. <bu _{0,6} > (ta _{0,6} .qa _{0,6}) _{0,9}									-0,9				-62,1
☞ b. <bu _{0,6} > (ta _{0,6} .qa _{0,6}) _{0,9}						-1		-0,8					-66,0

(56) [[du-ʔya:q-ad-qa] -ba]

	∞ *W<D	1.070 MAXAL	960 $\mu_{\mu\mu}$ * σ	960 *IM<: *#(<sigma>)	151 *ILH: *#(<V>:)	10 *V: #	140 IdFt(Hd)	70 DEPAL	69 *EVIMB	60 FTBIN	35 PARSE	10 ALIGN(Hd;L)	$\mathcal{H}$
① No MF and stress on the second foot													
a. <du _{0,8} >(ya _{0,8}) ₁ (qa _{0,8} d) ₁ qa _{0,8}				-1	-1						-1		-1.146,0
b. <du _{0,8} >(ya _{0,8}) ₁ (qá _{0,8} d) ₁ qa _{0,8}				-1							-1	-1	-1005,0
c. <du _{0,8} >(ya _{0,8} .qá _{0,8} d) ₁ qa _{0,8}		-1	-1					-1			-1		-2.135,0
② No MF and stress remains													
a. <du _{0,6} >(ya _{0,6}) _{0,9} (qá _{0,6} d) _{0,9} (qa _{0,6} .ba _{0,8}) ₁				-1					-1			-1	-1.039,0
b. <du _{0,6} >(ya _{0,6}) _{0,9} (qá _{0,6} d) _{0,9} (qa _{0,6} .ba _{0,8})				-1				-1				-1	-1.040,0
c. <du _{0,6} >(ya _{0,6}) _{0,9} (qa _{0,6} d) _{0,9} (qa _{0,6} .ba _{0,8}) ₁				-1	-1		-1		-1				-1.320,0
d. <du _{0,6} >(ya _{0,6} .qa _{0,6} d) _{0,9} (qa _{0,6} .bá _{0,8}) _{0,9}		-0,8	-1					-0,8	-1			-1	-1.951,0

Recall from 4.2.2 that MAX- μ_{INI} has a weight of at least 860; complete deletion of an initial mora is hence not an option.

(57) [[qɑ:ʔiɖ] -ba]

	∞ *W _K D	MAXAL	$\sigma_{\mu\mu\mu}^*$	960	960	*IM _O :*#(σ)	151	*ILH:*#(V̇:)	10	*V:#	140	dFt(H _D)	70	DEPAL	69	*EVIMB	60	FTBIN	35	PARSE	10	ALIGN(H _D ;L)	$\mathcal{H}$
① No MF and stress on the second foot																							
a. (qá:0,8) ₁ (ʔí0,8d) ₁					-1	-1																	-1.111,0
☞ b. (qɑ:0,8) ₁ (ʔí0,8d) ₁					-1																-1		-970,0
c. (qɑ0,8.ʔí:0,8d) ₁			-1	-1					-1				-1										-2.110,0
② No MF and stress remains																							
☞ a. (qɑ:0,6) _{0,9} (ʔí0,6d) _{0,9} ba0,8					-1														-1	-1			-1.005,0
b. (qá:0,6) _{0,9} (ʔí0,6d) _{0,9} ba0,8					-1	-1					-1								-1				-1.286,0
c. (qɑ0,6.ʔí:0,6d) _{0,9} (bá0,8) _{0,9}			-0,8	-1									-0,8				-1				-1		-1.942,0

(58) [[qa:ʔid] -u]

	∞ *W _{KD}	M _{AXAL}	σ^{1111}	960 *IM _O :*#(σ)	151 *IL _H :*#(V̇):	10 *V: #	140 IdFt(H _D)	D _{EPAL}	69 *EvIMB	60 FtBIN	35 PARSE	10 ALIGN(H _D ;L)	$\mathcal{H}$
② MF, stress remains													
a. (qa: _{0,6}) _{0,9} (ʔi _{0,6} .dú _{0,8}) _{0,9}				-1					-0,9			-1	-1.032,1
b. (qá: _{0,6}) _{0,9} (ʔi _{0,6} .du _{0,8}) _{0,9}				-1	-1		-1		-0,9				-1.313,1
c. (qa _{0,6} .ʔi: _{0,6}) _{0,9} (dú _{0,8}) _{0,9}		-0,8						-0,8		-1		-1	-982,0
d. (qa _{0,6} .ʔi: _{0,6}) _{0,9} du _{0,8}		-0,8					-1	-0,8			-1		-1087,0
e. (qa _{0,6} .ʔi: _{0,6}) _{0,9} (dú: _{0,8}) _{0,9}		-0,8				-1		-1,8				-1	-1002,0

(59) [[q'a:] -mela]

	∞ *W _K D	1.070 MAXAL	960 * $\sigma^{\mu\mu\mu}$	960 *IM _O :*#(σ)	151 *ILH:*($\acute{V}$):	10 *V:##	140 dFt(H _D)	70 DEPAL	69 *EVIMB	60 FTBIN	35 PARSE	10 ALIGN(H _D ;L)	$\mathcal{H}$
① Main stress on only long vowel													
☞ a. (q'á:0,8) ₁				-1	-1	-1							-1.121,0
b. (q'á0,8) ₁		-1								-1			-1.130,0
② Stress shift to unmarked short vowel													
a. (q'á:0,6) _{0,9} (me0,8.la0,8) ₁				-1	-1				-1				-1.180,0
☞ b. (q'a:0,6) _{0,9} (me0,8.lá0,8) ₁				-1			-1		-1			-1	-1.179,0
c. (q'a0,6.me:0,8) _{0,9} (lá0,8) ₁	-1	-0,8					-1	-1		-1		-1	$-\infty$

Candidate (59-①-b) violates MAX- μ in addition to MAXAL(V- μ)– as is discussed above in subsection 5.2, this violation is not even necessary since the latter is decisive on its own in all contexts.

(60) [[loq'o:tf -ad -uwad] -u]

	∞ *W _K D	MAXAL 1.070	σ^* _{tuq} 960	*IM _O :*#(σ) 960	*IL _H :*#(V:) 151	*V:## 10	D _{FT} (H _D) 140	DEPAL 70	*EVIMB 69	FTBIN 60	PARSE 35	ALIGN(H _D :L) 10	$\mathcal{H}$
① No MF, main stress on second foot													
a. <lo _{0,8} > (qó _{0,8}) ₁ (tʃa _{0,8} .du _{0,8}) ₁ (wa _{0,8} d) ₁				-1	-1				-1				-1.180,0
☞ b. <lo _{0,8} > (qo _{0,8}) ₁ (tʃa _{0,8} .dú _{0,8}) ₁ (wa _{0,8} d) ₁				-1					-1			-1	-1.039,0
c. <lo _{0,8} > (qo _{0,8} .tʃá _{0,8}) ₁ (du _{0,8} .wa _{0,8} d) ₁		-1						-1					-1.140,0
② MF, main stress remains on second foot													
a. <lo _{0,6} > (qo _{0,6}) _{0,9} (tʃa _{0,6} .dú _{0,6}) _{0,9} (wa _{0,6} .du _{0,8}) ₁				-1					-1,9			-1	1.101,1
b. <lo _{0,6} > (qo _{0,6}) _{0,9} (tʃa _{0,6} .dú _{0,6}) _{0,9} (wa _{0,6} .du _{0,8}) ₁				-1				-0,8	-1			-1	1.095,0
c. <lo _{0,6} > (qo _{0,6} .tʃa _{0,6}) _{0,9} (du _{0,6} .wá _{0,6}) _{0,9} du _{0,8}		-0,8						-0,8	-0,9		-1	-1	-1.030,3
☞ d. <lo _{0,6} > (qo _{0,6} .tʃa _{0,6}) _{0,9} (du _{0,6} .wá _{0,6}) _{0,9} du _{0,8}		-0,8						-1,6			-1	-1	-1.024,2
e. <lo _{0,6} > (qo _{0,6} .tʃá _{0,6}) _{0,9} (du _{0,6} .wa _{0,6}) _{0,9} du _{0,8}		-0,8					-1	-1,6			-1		-1.154,2

(61) [[s'i] -p^hila]

	∞ *W _{KD}	1.070 MAXAL	960 * $\sigma_{\mu\mu\mu}$	960 *IM _O :*#(σ)	151 *IL _H :*#($\acute{V}$:)	10 *V:#	140 IdFt(H _D)	70 DEPAL	69 *EVIMB	60 FTBIN	35 PARSE	10 ALIGN(H _D ;L)	$\mathcal{H}$
① H-docking but no IL													
☞ a. (s'í _{0,8}) ₁										-1			-60,0
b. (s'í _{0,8}) ₁				-1	-1	-1		-1					-1.191,0
② H has to move but no IL for Wd-Sfx													
a. (s'í _{0,6}) _{0,9} (p ^h í _{0,8} .la _{0,8}) ₁									-1	-1			-129,0
☞ b. (s'í _{0,6} .p ^h í _{0,8}) _{0,9} la _{0,8}									-0,9		-1		-97,1
c. (s'í _{0,6} .p ^h í _{0,8}) _{0,9} la _{0,8}								-1			-1		-105,0
d. (s'í _{0,6}) _{0,9} (p ^h í _{0,8} .la _{0,8}) ₁				-1	-1			-0,8	-1				-1.236,0
e. (s'í _{0,6}) _{0,9} (p ^h í _{0,8} .lá _{0,8}) ₁				-1			-1	-0,8	-1			-1	-1.235,0

The comparison between (61-②-a) and (61-②-b) reveals a potential misprediction of the system: For an input [[CV] -CVCV:(...)], an output *(C $\acute{V}$) (CV.CV:) (...) is predicted to be optimal. This follows since *EVENIAMB out-weighs ||FTBIN||. I am not aware of any such example. In case the output for such an input structure is indeed (CV.C $\acute{V}$) (CV:) (...), the inclusion of a more specific ||FTBIN|| constraint with a weight of minimally 53 demanding that initial feet must be binary would easily make the correct prediction.